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Chapter I: Introduction

Computer Vision is one of the most important fields of Artificial Intelligence, focusing on enabling computers to understand and interpret visual information obtained from images and videos. In recent years, the rapid development of deep learning techniques has significantly improved the performance of computer vision systems, making them suitable for real-world applications such as medical diagnosis, surveillance systems, industrial inspection and autonomous driving.

Among the various applications of Computer Vision, autonomous driving has attracted considerable attention from both researchers and industry. An autonomous vehicle must be capable of understanding its surrounding environment, identifying road structures, recognizing obstacles and making decisions based on visual information collected through cameras and sensors. For this reason, road scene understanding represents a fundamental component in the development of intelligent transportation systems.

In this project, the focus is placed on road scene classification and semantic segmentation. The classification task aims to determine the type of driving environment present in a given image, while semantic segmentation aims to identify and localize different regions within the image at the pixel level. Combining these two tasks provides a more complete understanding of road environments and contributes to the development of more reliable autonomous driving systems.

For the implementation of the classification model, the BDD100K dataset was selected because it contains a large collection of real-world driving images recorded under different weather conditions, lighting environments and road scenarios. From the available classes, three representative categories were selected: City Street, Highway and Residential. These categories were chosen because they represent common driving environments and provide a suitable level of difficulty for image classification.

The classification model was developed using the MobileNetV3Large convolutional neural network architecture. Instead of training a deep neural network from scratch, transfer learning techniques were employed by utilizing pretrained ImageNet weights. Furthermore, fine-tuning strategies and data augmentation techniques were applied in order to improve the generalization capability of the model and increase classification performance.

In addition to classification, semantic segmentation techniques were also investigated in order to identify important road regions within the images. By combining classification and segmentation, the proposed system is capable of providing both global scene understanding and detailed spatial information about the driving environment.

The main objective of this project is to develop a robust road scene understanding system capable of classifying different road environments while also providing semantic segmentation results. The performance of the proposed approach is evaluated through accuracy metrics, confusion matrices, Grad-CAM visualizations and additional analysis techniques in order to better understand the behaviour of the trained models.

Chapter II: Dataset Description

2.1 BDD100K Dataset

One of the most important steps in developing a deep learning model is the selection of an appropriate dataset. Since the objective of this project is road scene understanding for autonomous driving, a dataset containing real-world driving scenarios was required. For this purpose, the Berkeley DeepDrive 100K (BDD100K) dataset was selected. BDD100K is one of the largest publicly available datasets for autonomous driving research. It contains more than 100,000 images collected from dashboard cameras installed on vehicles driving in different environments. The images were captured under a wide variety of conditions, including daytime and nighttime scenarios, different weather conditions, urban areas, highways and residential neighborhoods.

One of the main advantages of BDD100K is the diversity of its data. Unlike small academic datasets, BDD100K contains images recorded in real traffic situations, making it particularly suitable for training computer vision models that are expected to generalize well to unseen environments. Furthermore, the dataset provides annotations for several computer vision tasks such as object detection, lane detection, semantic segmentation and scene classification. Considering the objective of this project, BDD100K provides a realistic benchmark for evaluating the performance of deep learning models in road scene classification tasks.

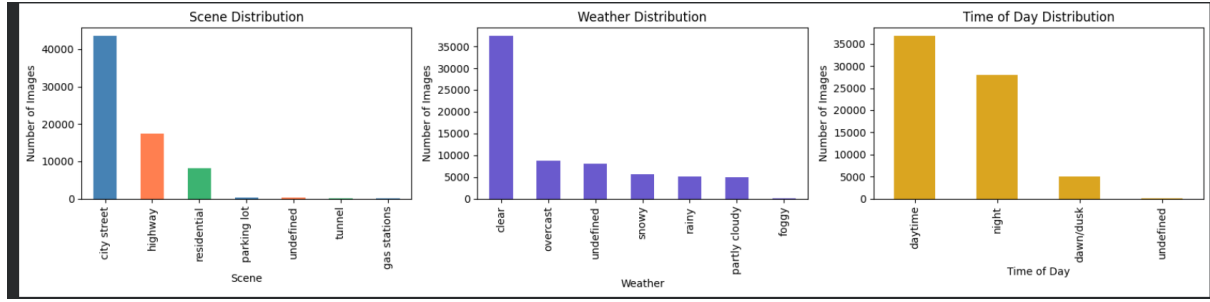


Figure 1: Distribution of Scene Categories

2.2 Selected Classes

Although BDD100K contains multiple scene categories and environmental attributes, only three classes were selected for the classification task performed in this project. The selected classes are:

- City Street
- Highway
- Residential

These categories were chosen because they represent three of the most common driving environments encountered by autonomous vehicles. Furthermore, they present different visual characteristics that allow the model to learn meaningful features while still providing a challenging classification problem. The City Street category contains images captured in urban environments characterized by traffic lights, intersections, commercial buildings and dense traffic activity. The Highway category includes roads designed for high-speed travel, typically containing multiple lanes, limited intersections and relatively structured traffic flow. The Residential category consists of roads located within residential areas. These scenes generally contain houses, parked vehicles, trees and lower traffic density compared to city streets.

Although these categories appear visually distinct, some overlap exists between them. For example, certain residential streets may resemble urban streets, making classification more challenging. This characteristic contributes to the complexity of the problem and provides a more realistic evaluation of model performance.

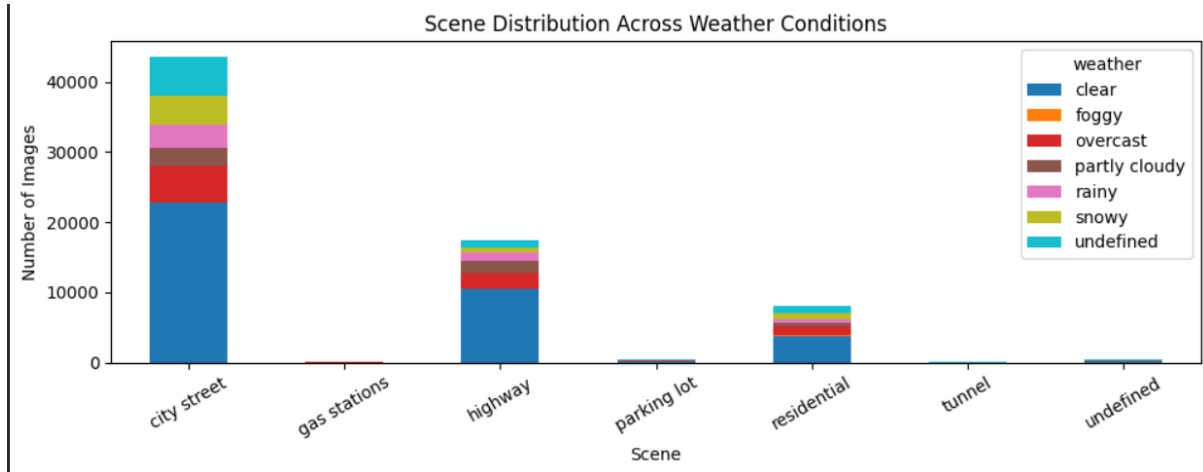


Figure 2: Distribution of Scene Categories Across Different Weather Conditions.

2.3 Dataset Preparation and Balancing

Before training the neural network, several preprocessing operations were performed on the dataset. The first step consisted of extracting only the images belonging to the selected classes. Images associated with other scene categories were excluded from the study. After the filtering process, the distribution of samples among the classes was analyzed. Since class imbalance can negatively affect the performance of machine learning models, a balancing strategy was applied to ensure that all classes contained an equal number of samples. This prevents the model from becoming biased toward a particular class during training.

The resulting dataset was then randomly shuffled and divided into three subsets:

- Training Set
- Validation Set
- Test Set

The training set was used to optimize the parameters of the neural network, while the validation set was used to monitor the learning process and reduce the risk of overfitting. Finally, the test set was reserved for the final evaluation of the model and was not used during training. By applying these preparation steps, a balanced and representative dataset was obtained, providing a reliable basis for training and evaluating the proposed classification model.

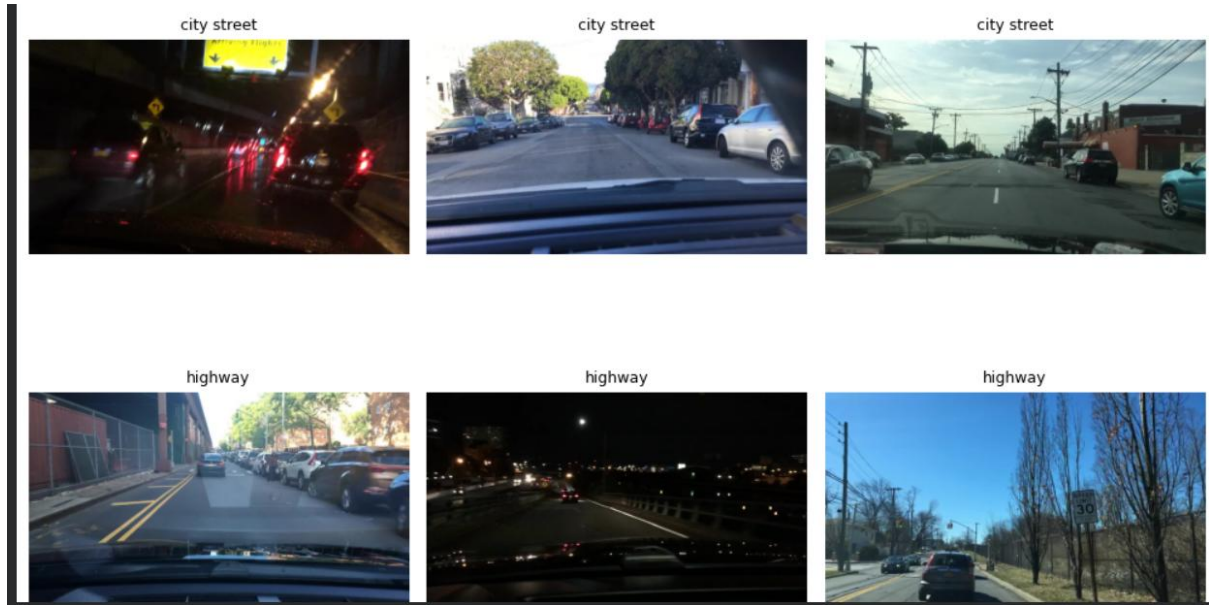


Figure 3: Representative Examples of the Three Selected Scene Classes

2.4 Image Preprocessing

Before training the classification model, several preprocessing operations were performed in order to improve data quality and ensure compatibility with the selected deep learning architecture. The first preprocessing step consisted of filtering the original BDD100K dataset and retaining only the images belonging to the three selected classes: City Street, Highway and Residential. Images associated with other scene categories were excluded from the analysis. After the filtering process, the class distribution was examined. Since deep learning models can be negatively affected by class imbalance, a balancing procedure was applied to ensure that all classes contained approximately the same number of samples. This step reduces the possibility of the model becoming biased toward a particular category.

The dataset was then randomly shuffled and divided into three subsets corresponding to training, validation and testing. This separation allows the model to be trained and evaluated on independent data, providing a more reliable estimate of its performance. All images were resized to a fixed resolution of 224×224 pixels, which corresponds to the input requirements of the MobileNetV3Large architecture. Furthermore, pixel values were preprocessed using the normalization procedure provided by the MobileNetV3 preprocessing function. This operation transforms the images into a format that is consistent with the ImageNet pretrained weights used during transfer learning. In order to improve the generalization capability of the model and reduce overfitting, several data augmentation techniques were applied during training. These transformations include random horizontal

flips, small rotations, zoom operations and brightness variations. Data augmentation artificially increases the variability of the training set and exposes the neural network to a broader range of visual conditions.

As a result of these preprocessing steps, the dataset becomes more balanced, consistent and suitable for training a robust road scene classification model.

Chapter III: Methodology

3.1 Transfer Learning

One of the main challenges in deep learning is the large amount of data and computational resources required to train deep neural networks from scratch. Although the BDD100K dataset contains a considerable number of images, training a convolutional neural network from the beginning would require significant computational power and training time. To address this limitation, a transfer learning approach was adopted. Transfer learning is a technique in which a model previously trained on a large dataset is reused for a new task. Instead of learning all visual features from the beginning, the model can exploit the knowledge already acquired during pretraining.

In this project, the MobileNetV3Large architecture was initialized using weights pretrained on the ImageNet dataset. Since ImageNet contains millions of images belonging to thousands of categories, the pretrained model has already learned general visual features such as edges, textures, shapes and object patterns. These features can then be adapted to the road scene classification task through additional training on the selected BDD100K classes. The use of transfer learning significantly reduces training time while improving classification performance and model generalization.

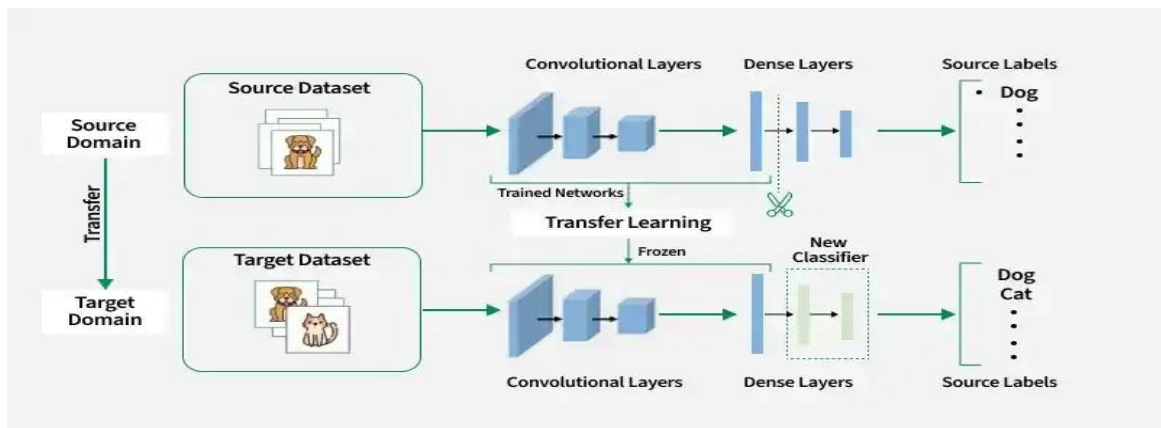


Figure 4: Transfer Learning

3.2 Deep Learning Architectures for Classification

In order to investigate the effectiveness of different transfer learning approaches, several convolutional neural network architectures were evaluated. The primary model employed in this study was MobileNetV3Large. Additionally, EfficientNetV2S, ConvNeXtSmall and VGG19 were trained and evaluated to provide a comparative analysis of classification performance.

3.2.1 MobileNetV3Large

For the classification task, the MobileNetV3Large convolutional neural network architecture was selected. The choice of this architecture was mainly motivated by its ability to provide a good balance between classification accuracy and computational efficiency. Since road scene classification requires the extraction of meaningful visual features from a large number of images, it was important to use a model capable of learning complex patterns while maintaining a reasonable training time. MobileNetV3Large satisfies these requirements and has become one of the most widely used architectures for image classification tasks.

The network receives as input RGB images resized to 224×224 pixels. The first operation performed by the model is a standard convolution layer whose objective is to extract low-level visual features from the image. At this stage, the network learns simple patterns such as edges, corners, textures and color transitions. Although these features appear relatively simple, they represent the foundation upon which more complex visual representations are built in the subsequent layers of the network.

After the initial convolution layer, the image is processed through a sequence of bottleneck blocks, which constitute the core of the MobileNetV3Large architecture. Unlike conventional convolutional neural networks, MobileNetV3 employs an inverted residual structure. In practice, this means that the feature maps are first expanded into a higher-dimensional space, then processed through depthwise convolutions and finally projected back into a lower-dimensional representation. This approach allows the network to learn richer feature representations while maintaining a relatively small number of parameters.

The expansion phase is performed using a 1×1 convolution that increases the number of channels present in the feature map. By increasing the dimensionality of the representation, the network is able to capture more detailed information about the visual content of the image. Once the expansion step is completed, a depthwise convolution is applied. In a traditional convolution operation, each filter processes all available channels

simultaneously. In contrast, depthwise convolution applies a separate filter to each channel independently. This significantly reduces the computational cost of the model and represents one of the main reasons why MobileNet architectures are considerably more efficient than conventional convolutional neural networks. Following the depthwise convolution, a projection layer is employed to reduce the dimensionality of the feature maps. This layer compresses the information generated during the expansion phase while preserving the most relevant visual characteristics. Consequently, the model is able to maintain a compact representation without sacrificing classification performance.

Another important component of MobileNetV3Large is the use of inverted residual connections. These connections facilitate the propagation of gradients during training and contribute to a more stable optimization process. Furthermore, they help mitigate the vanishing gradient problem, which often affects very deep neural networks. By allowing information to flow more efficiently across layers, residual connections improve both training stability and overall model performance.

Several bottleneck blocks also incorporate Squeeze-and-Excitation (SE) modules. The objective of these modules is to enable the network to determine which feature channels are more important for the classification task. First, global information is extracted from the feature maps through a global pooling operation. Subsequently, a small neural network calculates importance weights for each channel. Finally, these weights are used to emphasize informative channels and suppress less relevant ones. This attention mechanism allows the model to focus on the most discriminative visual features and contributes to improved classification accuracy.

In addition to the structural modifications described above, MobileNetV3Large introduces the Hard-Swish activation function. Compared to the traditional ReLU activation function, Hard-Swish provides smoother gradient propagation and improves optimization during training. At the same time, it maintains a low computational cost, making it particularly suitable for lightweight architectures designed for efficient inference.

Near the end of the network, the extracted feature maps are processed using a Global Average Pooling layer. Instead of flattening the entire feature tensor into a very large vector, the network computes the average value of each feature map. This operation substantially reduces the number of trainable parameters, decreases the risk of overfitting and improves the

generalization capability of the model. The resulting feature vector contains a compact summary of the most relevant information extracted from the image.

Since the original MobileNetV3Large architecture was pretrained on the ImageNet dataset, its final classification layer was removed and replaced by a custom classification head adapted to the requirements of this project. The custom head consists of a fully connected dense layer followed by a dropout layer used to reduce overfitting. Finally, a dense output layer containing three neurons was introduced, corresponding to the three scene categories considered in this study: City Street, Highway and Residential. A Softmax activation function was applied to transform the outputs into probability values, allowing the model to determine the most likely class for each input image.

Overall, the resulting architecture combines the feature extraction capabilities of the pretrained MobileNetV3Large backbone with a task-specific classification head designed for road scene classification. This combination allows the model to leverage previously learned visual representations while adapting effectively to the characteristics of the selected BDD100K dataset.

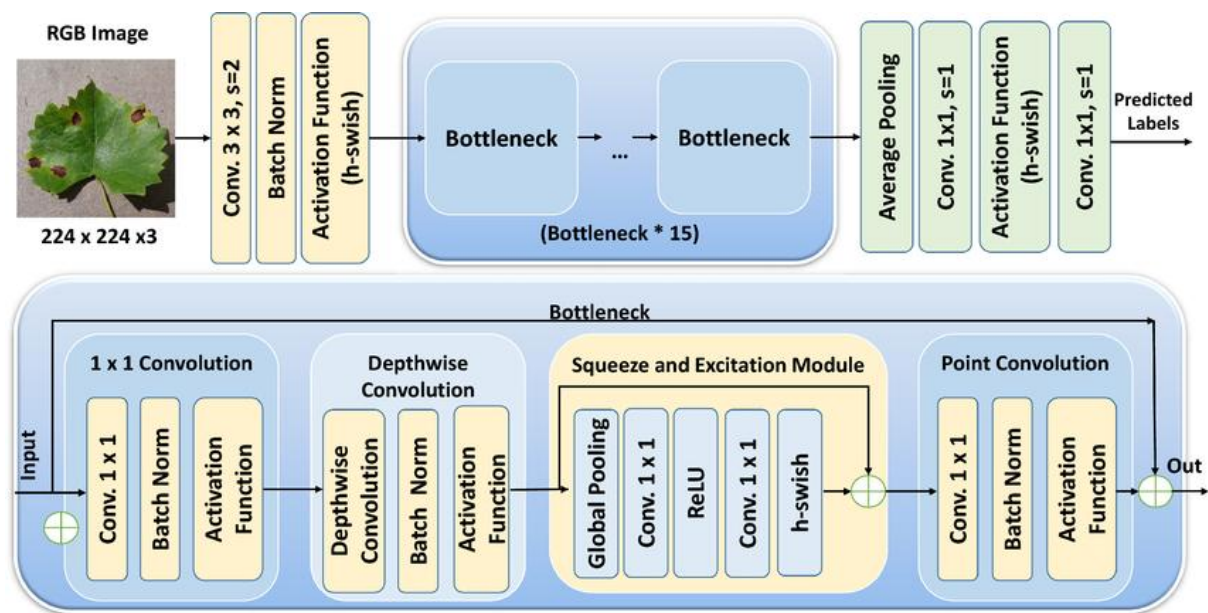


Figure 5: Architecture of the MobileNetV3Large Network

Figure presents the overall structure of the MobileNetV3Large architecture used in this project. As can be observed, the network begins with an initial convolution layer followed by a sequence of bottleneck blocks responsible for progressively extracting more complex visual features from the input image. Each bottleneck block combines expansion layers, depthwise convolutions, squeeze-and-excitation modules and projection layers. These

components work together to reduce computational complexity while preserving the ability of the network to learn highly discriminative feature representations. Finally, the extracted features are aggregated through a global average pooling operation before being passed to the classification layers responsible for generating the final prediction.

3.2.2 EfficientNetV2S

In addition to MobileNetV3Large, the EfficientNetV2S architecture was also evaluated as part of the experimental study. EfficientNetV2 belongs to a family of convolutional neural networks designed to achieve high classification accuracy while maintaining efficient training and inference times. The architecture represents an evolution of the original EfficientNet models and introduces several improvements aimed at accelerating training without sacrificing predictive performance.

One of the key characteristics of EfficientNetV2S is the use of a compound scaling strategy, which balances network depth, width and image resolution in a systematic manner. This approach allows the model to utilize computational resources more effectively while maintaining a relatively compact architecture. Furthermore, EfficientNetV2 incorporates fused convolutional blocks and optimized training procedures that contribute to improved efficiency compared to previous generations of deep neural networks.

In this project, EfficientNetV2S was employed as a comparative model in order to evaluate whether a larger and more sophisticated architecture could provide additional improvements in road scene classification performance. The model was initialized with ImageNet pretrained weights and adapted to the three selected scene categories through transfer learning techniques. The resulting performance was subsequently compared with the MobileNetV3Large baseline in order to assess the advantages and limitations of both approaches.

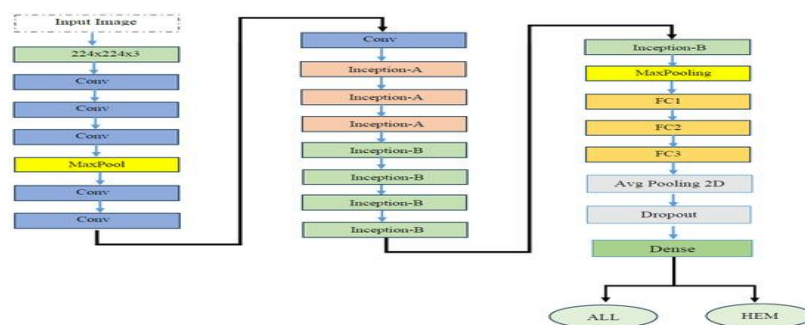


Figure 6: Architecture of the EfficientNetV2S Deep Learning Model

3.2.3 ConvNeXtSmall

Another architecture considered in this study is ConvNeXtSmall. Although convolutional neural networks have dominated computer vision research for many years, recent advances in Vision Transformers have introduced new architectural design principles. ConvNeXt was developed with the objective of incorporating several ideas inspired by transformer-based models while preserving the simplicity and efficiency of convolutional operations. The ConvNeXt architecture modernizes traditional convolutional neural networks through the use of larger kernel sizes, improved normalization strategies and updated network design choices. As a result, ConvNeXt achieves performance levels comparable to transformer-based architectures while maintaining the advantages of convolutional feature extraction. These characteristics make ConvNeXt an attractive candidate for image classification tasks involving complex visual patterns.

Within the context of this project, ConvNeXtSmall was included as an additional benchmark architecture. By evaluating its performance on the BDD100K dataset, it becomes possible to determine whether modernized convolutional designs offer significant advantages for road scene understanding. The model was trained using transfer learning and subsequently compared against the other architectures considered in this work.

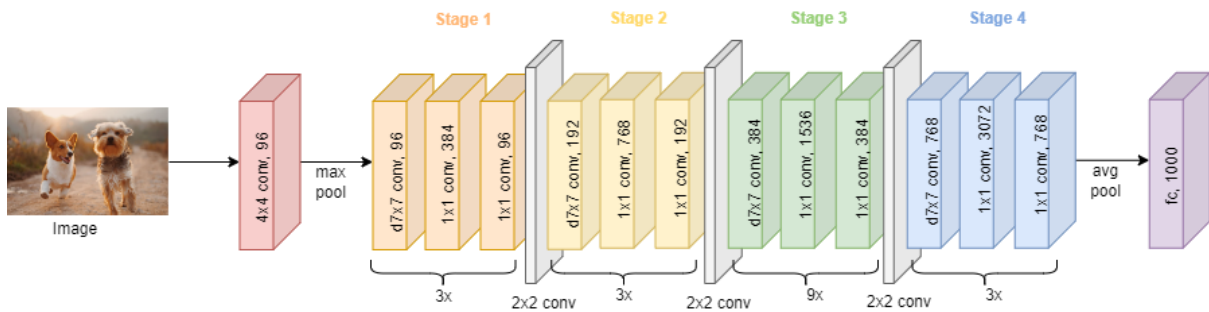


Figure 7: Architecture of the ConvNeXtSmall Deep Learning Model

3.2.4 VGG19

The VGG19 architecture was selected as a classical deep learning baseline for comparison purposes. Developed by the Visual Geometry Group (VGG) at the University of Oxford, VGG19 is one of the most influential convolutional neural network architectures in the history of computer vision. Despite being introduced several years before modern architectures such as MobileNetV3 and EfficientNetV2, VGG19 remains widely used as a reference model due to its simple and intuitive design.

The architecture consists of a deep sequence of convolutional layers using small 3×3 filters, followed by fully connected classification layers. Although VGG19 achieves strong classification performance, it contains a significantly larger number of parameters compared to more recent architectures. Consequently, it generally requires higher computational resources and longer training times.

In this project, VGG19 was utilized as a benchmark model to provide a comparison between traditional deep convolutional networks and more modern architectures optimized for efficiency. By comparing its classification performance with MobileNetV3Large, EfficientNetV2S and ConvNeXtSmall, it becomes possible to evaluate the extent to which recent architectural innovations contribute to improved road scene classification performance.

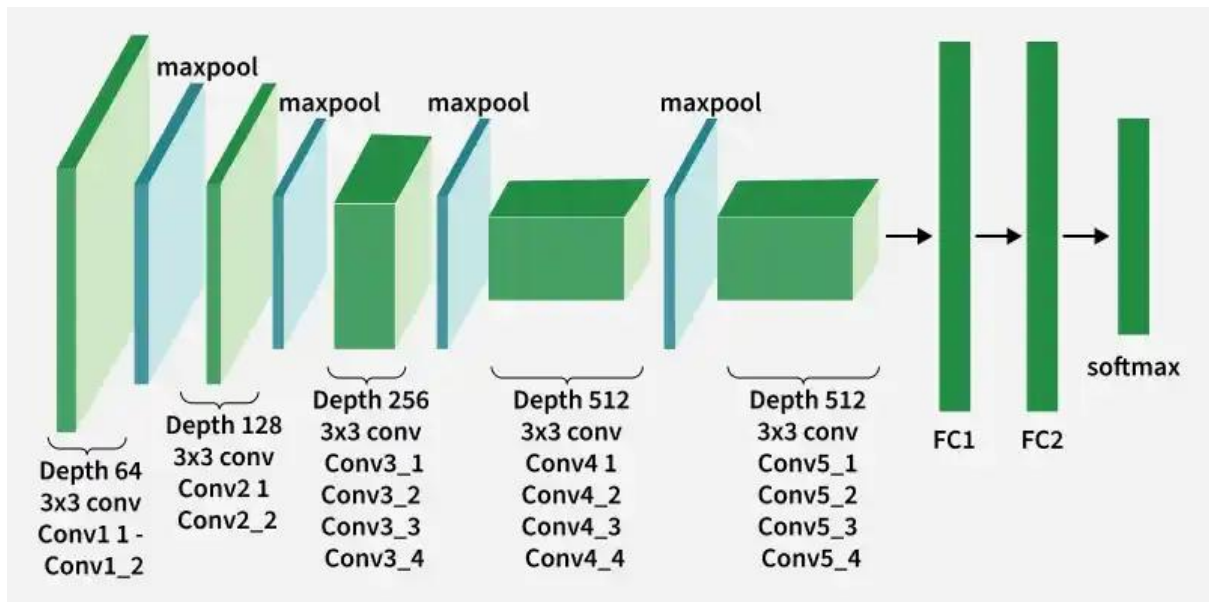


Figure 8: Architecture of the VGG19 Convolutional Neural Network

3.3 Fine-Tuning Strategy

As previously discussed, the MobileNetV3Large architecture used in this project was initialized with weights pretrained on the ImageNet dataset. Although these pretrained weights provide a strong starting point for image classification tasks, the visual characteristics of ImageNet images differ significantly from those encountered in autonomous driving scenarios. Consequently, additional adaptation of the network was required in order to improve its ability to recognize and distinguish between road scene categories.

The training procedure adopted in this project was divided into two distinct phases. During the first phase, the convolutional backbone of MobileNetV3Large was frozen and

only the newly introduced classification layers were allowed to update their parameters. In practice, this means that the feature extraction capabilities learned during pretraining remained unchanged, while the classification head learned how to associate the extracted features with the three target classes considered in this study, namely City Street, Highway and Residential.

This initial training stage plays an important role because it allows the newly added layers to adapt to the specific classification task without modifying the pretrained representations. Since the ImageNet weights already contain valuable information about general visual structures such as edges, textures, shapes and object patterns, freezing the backbone helps preserve this knowledge and prevents the network from losing useful information during the early stages of training.

After the classification layers reached a satisfactory level of convergence, a fine-tuning procedure was applied. During this phase, a subset of the final layers of the MobileNetV3Large backbone was unfrozen and allowed to participate in the optimization process. The objective of this step was to adapt the high-level feature representations of the network to the characteristics of road scene images contained in the BDD100K dataset.

The decision to unfreeze only the final layers rather than the entire network was motivated by the fact that the lower layers of a convolutional neural network generally learn universal visual features that remain useful across different image domains. In contrast, the deeper layers are responsible for learning more task-specific representations. Therefore, by fine-tuning only the upper portion of the network, it becomes possible to adapt the model to the target dataset while maintaining the robustness of the previously learned low-level features.

Another important aspect of the fine-tuning process is the selection of an appropriate learning rate. A significantly smaller learning rate was employed during this stage compared to the initial training phase. This choice was made to ensure that the pretrained weights were updated gradually and that large parameter modifications did not destroy the information acquired during pretraining. In other words, the network was encouraged to refine its existing knowledge rather than relearn all visual features from the beginning.

The effectiveness of the adopted fine-tuning strategy can be observed through the evolution of the training and validation curves presented in the results section. After the transition from feature extraction to fine-tuning, both training accuracy and validation

accuracy continued to improve, indicating that the network successfully adapted to the road scene classification problem. Furthermore, the final model achieved a noticeable increase in classification performance compared to the initial training stage, demonstrating the importance of fine-tuning when applying transfer learning techniques to specialized computer vision tasks.

Overall, the combination of transfer learning and fine-tuning allowed the model to exploit the advantages of pretrained ImageNet features while simultaneously learning the visual characteristics that distinguish urban streets, highways and residential roads. This strategy contributed significantly to the final performance obtained by the classification system.

3.4 Data Augmentation

One of the most common challenges encountered when training deep learning models is the phenomenon of overfitting. This problem occurs when a neural network becomes excessively specialized on the training data and consequently loses its ability to generalize effectively to unseen images. Although the BDD100K dataset contains a large number of road scene images, the subset selected for this project is considerably smaller and therefore more susceptible to overfitting. Furthermore, autonomous driving environments are characterized by a high degree of variability, including changes in lighting conditions, weather conditions, camera viewpoints and surrounding objects. For this reason, additional measures were required to increase the diversity of the training data and improve the robustness of the proposed classification model.

To address this issue, data augmentation techniques were employed during the training phase. Data augmentation is a preprocessing strategy that artificially increases the variability of the training dataset by generating modified versions of existing images. Unlike traditional data collection methods, which require acquiring additional images, augmentation allows new training samples to be produced directly from the available dataset through the application of random transformations. An important characteristic of these transformations is that they preserve the semantic meaning of the image. In other words, although the visual appearance of the image changes, its class label remains unchanged.

Several augmentation techniques were incorporated into the training pipeline. One of the transformations applied was random horizontal flipping. This operation mirrors the image along the vertical axis, producing an alternative representation of the same road scene. Since

roads, vehicles and surrounding structures can appear in different positions within an image, horizontal flipping helps the model become less sensitive to the spatial location of visual features.

In addition to horizontal flipping, small rotation transformations were introduced. During real-world driving scenarios, camera orientation may vary slightly due to vehicle movement, road inclination or sensor positioning. By exposing the neural network to rotated versions of the original images, the model becomes more robust to these natural variations and learns to recognize relevant visual patterns regardless of small geometric changes.

Zoom operations were also incorporated as part of the augmentation process. These transformations simulate situations in which objects appear at different distances from the camera. In autonomous driving environments, vehicles, buildings and road structures may occupy different portions of the image depending on their relative position. By applying random zoom transformations, the model learns to recognize scene characteristics at multiple scales, improving its ability to generalize to diverse driving situations.

Another important augmentation technique used in this project involves brightness variations. Road scenes may be recorded under highly different illumination conditions, including bright daylight, cloudy weather, dawn, dusk and nighttime environments. As a result, the same road category may exhibit substantial differences in visual appearance. Introducing random brightness adjustments allows the neural network to become less dependent on specific lighting conditions and encourages the learning of more robust feature representations.

The application of these augmentation techniques effectively increases the diversity of the training data without requiring additional image collection. Consequently, the neural network is exposed to a broader range of visual scenarios and learns feature representations that are more invariant to common environmental changes. This contributes directly to reducing overfitting and improving the generalization capability of the model.

Furthermore, data augmentation plays a particularly important role in road scene classification because autonomous driving systems must operate reliably under a wide variety of real-world conditions. A model trained exclusively on the original images may perform well on the training set but struggle when confronted with unseen situations. By introducing controlled variability during training, augmentation encourages the network to focus on the

fundamental characteristics that distinguish City Street, Highway and Residential environments rather than memorizing specific visual details present in the training images.

Therefore, data augmentation represents a crucial component of the proposed methodology and contributes significantly to the overall performance and robustness of the classification system developed in this project.

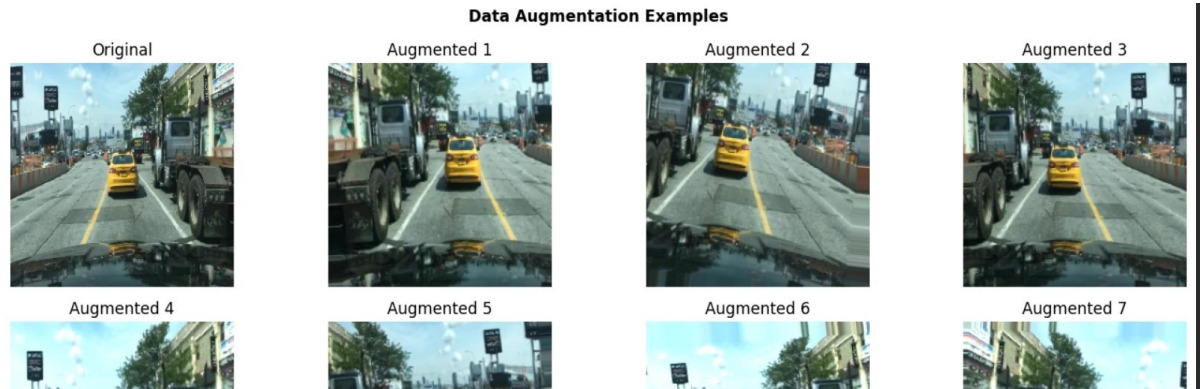


Figure 9: Data Augmentation Applied During Model Training

3.5 Model Training Configuration

After defining the architecture of the proposed classification model, the next step consisted of configuring the training process. The implementation was carried out using the TensorFlow deep learning framework together with the Keras high-level API. These libraries provide an efficient environment for developing, training and evaluating convolutional neural networks while offering a wide range of optimization and monitoring tools.

Prior to training, all images were resized to a resolution of 224×224 pixels. This image size was selected because it corresponds to the input requirements of the MobileNetV3Large architecture. Using a fixed image resolution ensures consistency throughout the dataset and allows the pretrained network to process all samples under the same conditions. Furthermore, resizing reduces memory consumption and computational requirements, making the training process more efficient while preserving the most important visual information contained in the images.

The classification problem addressed in this project involves three mutually exclusive classes: City Street, Highway and Residential. Since each image belongs to exactly one category, the categorical cross-entropy loss function was selected. This loss function is widely used in multi-class classification problems because it measures the difference between the predicted probability distribution generated by the network and the true class distribution.

During training, the optimization process attempts to minimize this loss value, encouraging the model to produce predictions that are increasingly closer to the correct labels.

To optimize the network parameters, the Adam optimization algorithm was employed. Adam combines the advantages of momentum-based optimization and adaptive learning rates, allowing the network to converge more efficiently compared to traditional gradient descent methods. Another important advantage of Adam is its ability to automatically adjust parameter updates during training, making it particularly suitable for deep learning applications involving large numbers of trainable parameters.

The dataset was divided into three independent subsets corresponding to training, validation and testing. The training set was used for updating the model parameters, while the validation set was employed to monitor the learning process and evaluate performance after each epoch. The use of a validation dataset is particularly important because it provides an indication of how well the model generalizes to unseen data. Finally, the testing set was reserved exclusively for the final evaluation stage and was not used during model optimization.

Several callback mechanisms were incorporated into the training pipeline in order to improve efficiency and reliability. Model checkpointing was used to automatically save the best-performing model according to validation performance. This ensures that the optimal set of weights is preserved even if later training epochs result in performance degradation. In addition, early stopping strategies were employed to prevent unnecessary training and reduce the risk of overfitting. By continuously monitoring validation performance, the training process can be interrupted when further improvements are no longer observed.

Overall, the adopted training configuration was designed to maximize classification performance while ensuring efficient use of computational resources and maintaining the ability of the model to generalize effectively to unseen road scene images.

3.6 Model Evaluation Metrics

Once the training process was completed, the performance of the proposed classification system was evaluated using several complementary metrics. Although overall classification accuracy provides a useful indication of model performance, relying on a single metric may not provide a complete understanding of the behaviour of the classifier. For this reason, additional evaluation measures were considered in order to perform a more comprehensive analysis.

The primary metric employed in this project is classification accuracy. Accuracy measures the proportion of correctly classified images relative to the total number of evaluated samples. Since the dataset was balanced across the three selected classes, accuracy provides a meaningful indicator of the overall effectiveness of the model. A higher accuracy value indicates that the classifier is capable of correctly distinguishing between different road scene categories.

However, accuracy alone does not reveal how the model performs on individual classes. For this reason, precision, recall and F1-score were also calculated. Precision measures the proportion of correctly classified samples among all predictions assigned to a given class. A high precision value indicates that the model rarely assigns incorrect labels to images belonging to other categories. Recall, on the other hand, measures the proportion of correctly identified samples relative to the total number of images belonging to a particular class. This metric provides insight into the ability of the model to detect all relevant instances of a given category.

The F1-score combines precision and recall into a single metric by computing their harmonic mean. This measure is particularly useful when comparing classification performance across multiple classes because it provides a balanced evaluation of both false positives and false negatives. Consequently, the F1-score offers a more detailed understanding of model behaviour than accuracy alone.

In addition to these numerical metrics, confusion matrices were generated to visualize the classification results. The confusion matrix provides a detailed breakdown of correct and incorrect predictions for each class and allows the identification of common classification errors. By examining the confusion matrix, it becomes possible to determine which road scene categories are frequently confused by the model and to analyze the underlying causes of these errors.

The combination of accuracy, precision, recall, F1-score and confusion matrices provides a comprehensive framework for evaluating the effectiveness of the proposed classification system. Together, these metrics offer both quantitative and qualitative insights into the performance of the trained neural network.

3.7 Grad-CAM Visualization

Although performance metrics provide valuable information regarding classification accuracy, they do not explain how the neural network reaches its decisions. Deep learning models are often considered "black boxes" because they are capable of producing highly accurate predictions while offering limited insight into the internal reasoning process. For this reason, interpretability techniques have become increasingly important in modern computer vision applications.

In this project, Gradient-weighted Class Activation Mapping (Grad-CAM) was employed to improve the interpretability of the proposed classification model. Grad-CAM is a visualization technique that highlights the image regions most influential in generating a particular prediction. Rather than simply indicating the predicted class, Grad-CAM provides visual evidence regarding the areas of the image that contributed most strongly to the decision-making process of the neural network.

The technique operates by analyzing the gradients flowing through the final convolutional layers of the network. These gradients are used to estimate the importance of different feature maps with respect to a specific class prediction. The resulting information is then transformed into a heatmap representing the relative importance of different image regions. Areas displayed with warmer colors correspond to regions that have a stronger influence on the prediction, while cooler colors indicate regions with lower importance.

By superimposing the generated heatmaps onto the original images, it becomes possible to visually inspect the attention of the network. This analysis helps determine whether the classifier is focusing on meaningful road-related structures or whether it is relying on irrelevant background information. Such verification is particularly important in safety-critical applications such as autonomous driving, where incorrect reasoning could potentially lead to unreliable decisions.

The Grad-CAM results obtained in this project revealed that the MobileNetV3Large model primarily focuses on road surfaces, lane markings, vehicles, intersections and surrounding urban structures. These observations indicate that the network successfully learned relevant visual features associated with the different scene categories. Furthermore, the visualizations provide additional confidence that the model is making decisions based on meaningful semantic information rather than random image patterns.

Beyond improving interpretability, Grad-CAM also serves as a valuable diagnostic tool. By examining both correctly classified and misclassified samples, it becomes possible to better understand the strengths and limitations of the model. Consequently, Grad-CAM complements the numerical evaluation metrics and contributes to a more complete assessment of the proposed road scene classification system.

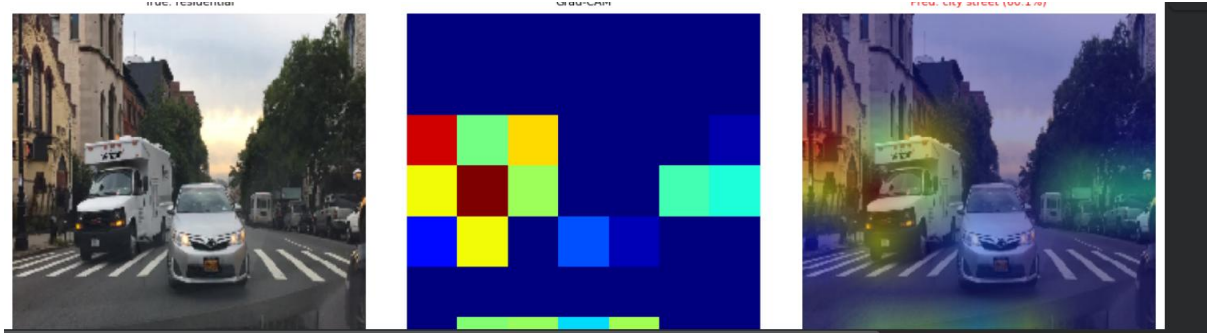


Figure 10: Grad-CAM Visualization Showing the Regions

Chapter IV: Performance Evaluation

4.1 Training Performance Analysis

The first step in evaluating the effectiveness of the proposed classification system consists of analyzing the behaviour of the model during the training process. Figure X presents the evolution of both training and validation accuracy together with the corresponding loss values obtained during the optimization of the MobileNetV3Large architecture.

As illustrated in the figure, the accuracy increases progressively throughout the training process. During the first epochs, a rapid improvement can be observed as the newly introduced classification layers learn to associate the extracted visual features with the corresponding road scene categories. As training progresses, the rate of improvement becomes more gradual, indicating that the model is approaching convergence.

The loss curves exhibit the expected behaviour for a successfully trained neural network. Both training loss and validation loss decrease steadily over time, demonstrating that the model continuously improves its ability to distinguish between City Street, Highway and Residential environments. Furthermore, no significant oscillations or abrupt increases are observed, suggesting that the optimization process remains stable throughout training.

An important aspect visible in Figure X is the transition to the fine-tuning stage after epoch 10. At this point, a subset of the pretrained MobileNetV3Large layers was unfrozen and allowed to update its parameters. Following this transition, additional improvements in both training and validation accuracy can be observed, while the corresponding loss values continue to decrease. This behaviour confirms that fine-tuning successfully adapts the pretrained ImageNet features to the characteristics of the BDD100K dataset.

Another noteworthy observation concerns the relatively small gap between training and validation performance. Throughout most of the training process, validation accuracy remains very close to training accuracy, indicating good generalization capability and limited overfitting. This behaviour suggests that the combination of transfer learning, data augmentation and regularization techniques contributes positively to the robustness of the proposed model.

Overall, the training curves demonstrate that the adopted methodology leads to a stable and effective learning process. The model converges successfully and achieves consistent performance across both training and validation datasets.

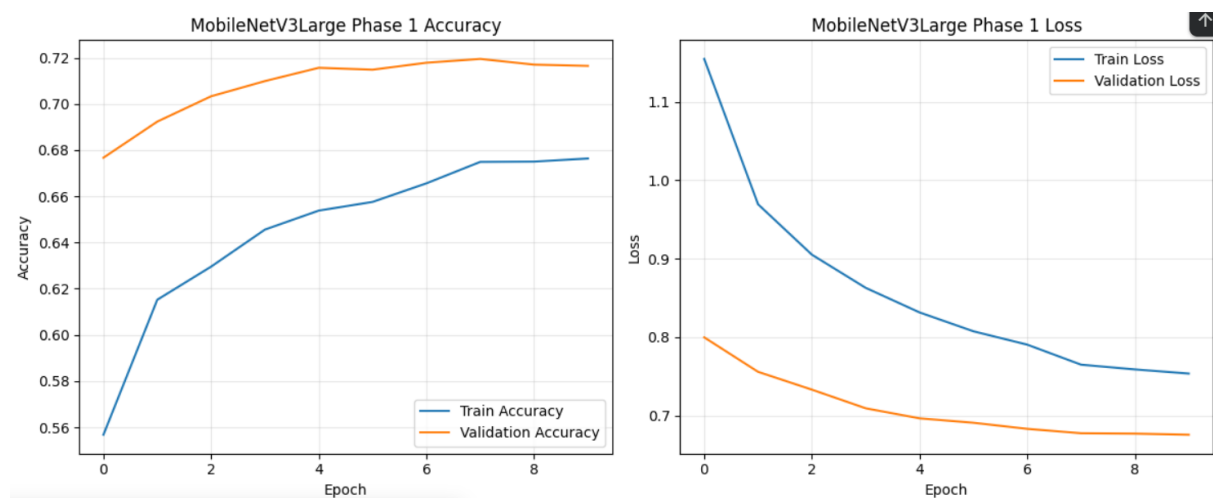


Figure 11: Training and Validation Accuracy and Loss Curves of MobileNetV3Large

4.2 Classification Performance

The proposed model achieved an overall test accuracy of 75.25% with a corresponding test loss of 0.6035. These results indicate that the network successfully learned discriminative representations capable of distinguishing between City Street, Highway and Residential environments. Considering the visual similarity that frequently exists between different road scene categories, the obtained accuracy demonstrates the effectiveness of the adopted transfer learning and fine-tuning strategy.

A more detailed analysis can be performed by examining the precision, recall and F1-score values for each class individually. The Highway category achieved the highest performance among all classes, with a precision of 0.78, a recall of 0.81 and an F1-score of 0.79. This suggests that highway scenes possess distinctive visual characteristics that allow the model to recognize them more consistently compared to the other categories.

The Residential class also produced strong results, achieving a precision of 0.76, a recall of 0.79 and an F1-score of 0.77. The relatively high recall indicates that most residential scenes were correctly identified by the model. This behaviour suggests that residential environments contain visual features that can be effectively learned by the neural network.

The most challenging category proved to be City Street, which obtained a precision of 0.72, a recall of 0.66 and an F1-score of 0.69. The lower recall value indicates that a portion of city street images were incorrectly classified as either Highway or Residential scenes. This outcome is expected because urban streets often share visual similarities with both categories, particularly in situations involving wide roads, varying traffic density and mixed urban infrastructure.

The per-class accuracy analysis further supports these observations. Highway achieved the highest class-specific accuracy of 80.92%, followed by Residential with 78.62%, while City Street obtained 66.20%. These results indicate that the majority of classification errors originate from the City Street category, highlighting the greater complexity associated with distinguishing urban road environments.

Overall, the classification metrics demonstrate that the proposed MobileNetV3Large model provides reliable performance across all three scene categories while maintaining balanced precision and recall values. The obtained results confirm the suitability of transfer learning techniques for road scene classification tasks and provide a solid foundation for the subsequent analyses presented in the following sections.

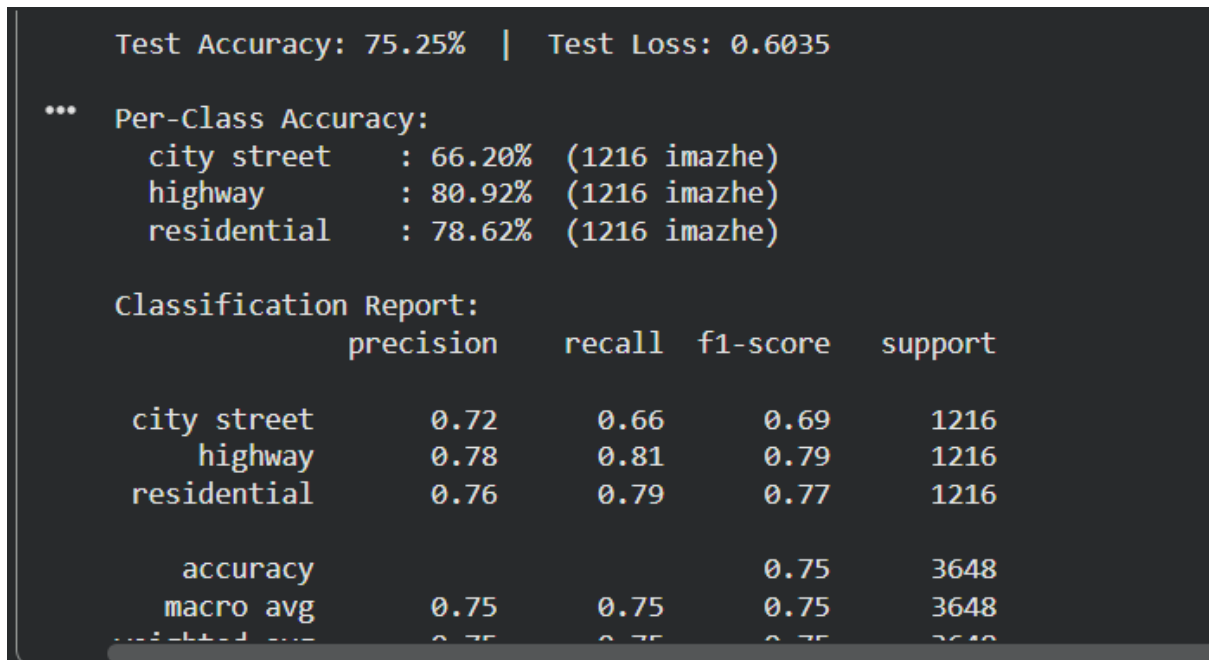


Figure 12: Classification Performance Metrics of MobileNetV3Large

4.3 Confusion Matrix Analysis

The first step in evaluating the effectiveness of the proposed classification system consists of analyzing the behaviour of the model during the training process. Figure 4.1 presents the evolution of both accuracy and loss values throughout the training procedure. These curves provide important information regarding the learning capability of the network and allow the identification of potential issues such as underfitting or overfitting.

As illustrated in the figure, the training accuracy increases steadily during the entire optimization process. Starting from approximately 56%, the model progressively improves its performance until reaching values above 72% at the end of training. A similar trend can be observed for the validation accuracy, which increases from approximately 68% to nearly 75%. The continuous improvement of both curves indicates that the network successfully learns meaningful visual representations capable of distinguishing between different road scene categories.

The loss curves exhibit the expected behaviour of a well-trained neural network. Both training loss and validation loss decrease consistently during optimization, demonstrating that prediction errors become progressively smaller as the learning process advances. Furthermore, no significant oscillations or instability can be observed, suggesting that the selected optimization parameters and learning strategy are appropriate for the considered classification task.

An important observation concerns the fine-tuning stage introduced after epoch 10. Following the unfreezing of a subset of pretrained layers, both accuracy curves continue to improve while the corresponding loss values decrease further. This behaviour confirms that the pretrained ImageNet features were successfully adapted to the characteristics of the BDD100K dataset. Consequently, the adopted transfer learning and fine-tuning strategy contributed significantly to the final performance achieved by the classification model.

Another noteworthy aspect is the relatively small difference between training and validation performance. Throughout most of the training process, both curves remain close to each other, indicating good generalization capability and limited overfitting. This observation suggests that the combination of transfer learning, dropout regularization and data augmentation effectively improves the robustness of the proposed model.

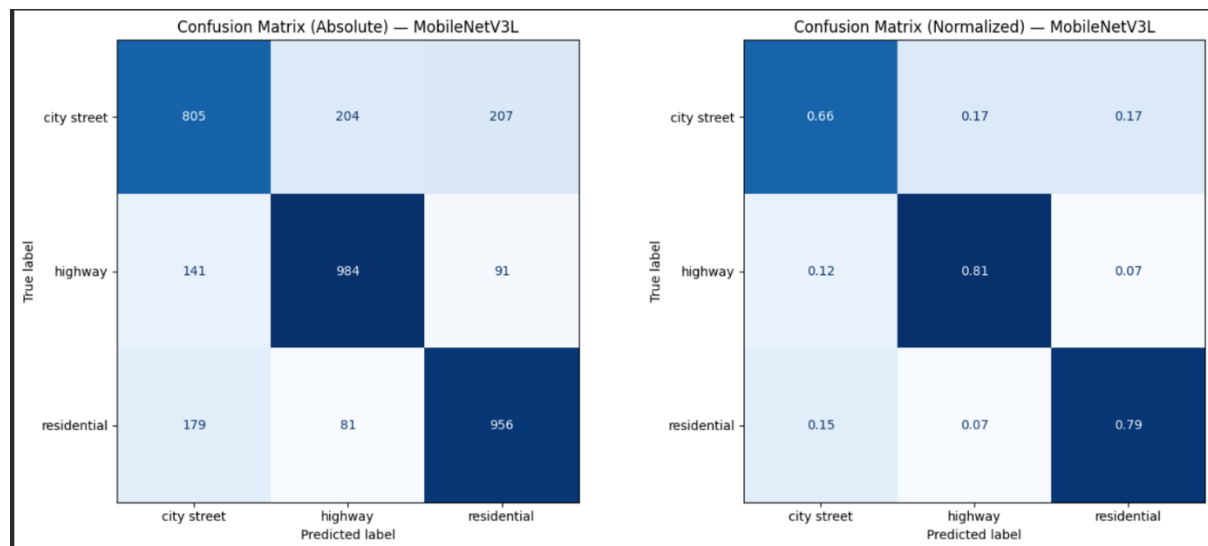


Figure 13: Confusion Matrix of MobileNetV3Large

4.4 ROC Curve Analysis

To further evaluate the discriminative capability of the proposed model, Receiver Operating Characteristic (ROC) curves were generated for each class. ROC analysis provides additional insight into classification performance by examining the relationship between the true positive rate and false positive rate across different decision thresholds.

As shown in Figure 13, all ROC curves remain significantly above the diagonal reference line corresponding to random classification. This observation indicates that the model possesses a strong ability to distinguish between positive and negative instances for each category.

The highest Area Under the Curve (AUC) value is achieved by the Highway class, reaching approximately 0.926. The Residential class follows closely with an AUC value of 0.917, while the City Street category achieves an AUC of 0.856. Although slightly lower, the City Street result still indicates strong discriminative performance.

The overall mean AUC value of approximately 0.900 confirms the effectiveness of the proposed classification framework and demonstrates that the model is capable of producing reliable predictions across all scene categories.

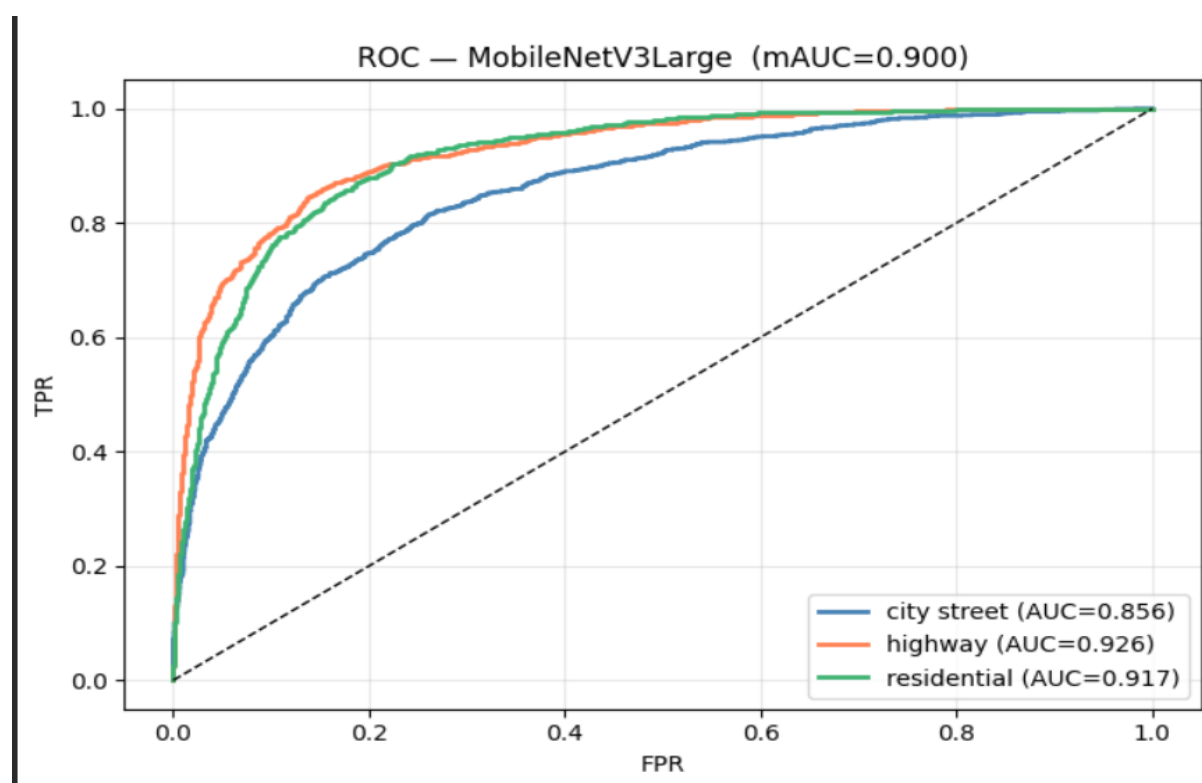


Figure 14: ROC Curves and AUC Scores

4.5 Grad-CAM Analysis

Although quantitative metrics such as accuracy, precision and recall provide valuable information regarding model performance, they do not explain how the neural network arrives at its decisions. For this reason, Grad-CAM visualizations were generated in order to investigate which image regions contribute most strongly to the classification process.

Figure 14 presents several examples of Grad-CAM activation maps produced by the MobileNetV3Large model. The visualizations reveal that the network primarily focuses on road surfaces, lane markings, vehicles, intersections and surrounding urban infrastructure when generating its predictions. These observations indicate that the model has learned to

identify semantically meaningful features associated with different road environments rather than relying on irrelevant image regions.

An important observation is that the highlighted regions frequently correspond to areas that humans would also consider relevant when distinguishing between road scene categories. In many cases, strong activations appear around vehicles, traffic lanes and road boundaries, suggesting that the network effectively captures contextual information describing the structure of the environment.

The Grad-CAM visualizations also provide insight into incorrect predictions. In one of the presented examples, a Residential scene is incorrectly classified as City Street. The activation map reveals that the network concentrates its attention on vehicles and dense urban structures located in the central region of the image. As a consequence, the model associates the scene with characteristics more commonly found in urban streets. Such observations help explain the reasoning process behind misclassifications and provide valuable information for future model improvements.

Overall, the Grad-CAM analysis confirms that the proposed classification system relies on meaningful visual features when generating predictions. The obtained visualizations increase the interpretability of the model and provide additional confidence regarding the validity of the learned feature representations.

4.6 Error Analysis

While the overall classification performance is satisfactory, a detailed investigation of incorrectly classified images provides valuable insight into the limitations of the proposed approach. Figure 4.5 presents several representative examples extracted from the test dataset where the model produced incorrect predictions.

A detailed examination of these images reveals that the majority of errors occur between the City Street and Highway categories. This behaviour is not unexpected because these environments often share common visual characteristics. Wide roads, multiple traffic lanes, moderate traffic density and urban infrastructure may cause certain city street scenes to resemble highway environments. Similarly, highways located near urban areas may contain visual elements that are frequently associated with city streets.

The first two examples shown in Figure 4.5 correspond to City Street images that were classified as Highway. In both cases, the road occupies a large portion of the image and

exhibits characteristics commonly observed in major traffic routes. Consequently, the model assigns a higher probability to the Highway category despite the correct label being City Street.

The third example illustrates the opposite situation, where a Highway scene is classified as City Street. The presence of multiple vehicles and dense traffic activity introduces visual patterns typically encountered in urban environments, leading the model toward an incorrect prediction.

Additional errors can also be observed between the Highway and Residential categories. These mistakes generally occur in situations where residential roads appear wider than usual or where highways contain strong visual elements such as trees, buildings or roadside structures that resemble suburban environments.

These observations indicate that most prediction errors are caused by the intrinsic visual similarity that exists between certain scene categories rather than by random failures of the neural network. Therefore, the obtained results highlight both the strengths and limitations of the proposed classification framework.

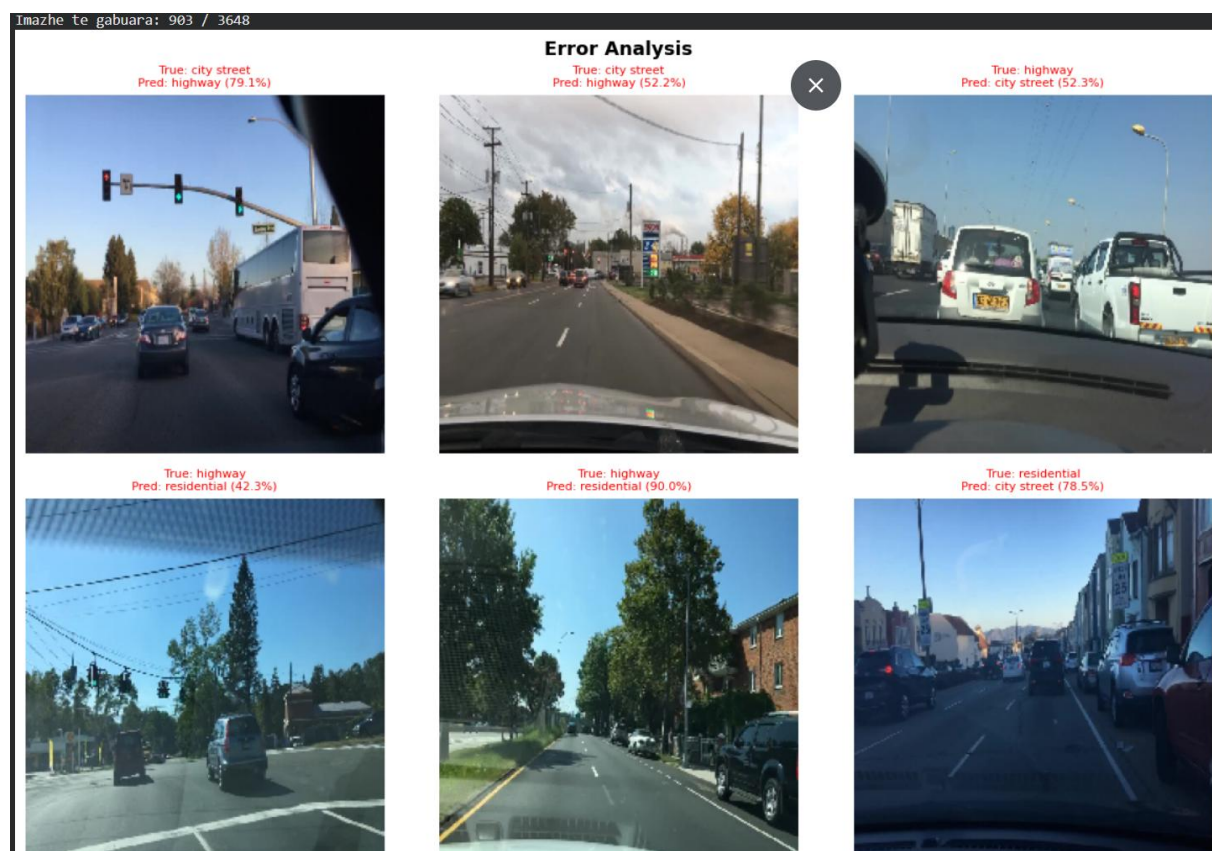


Figure 15: Examples of Misclassified Images Used for Error Analysis

4.7 Confidence Analysis

In addition to evaluating classification accuracy, it is also important to examine the confidence associated with individual predictions. Confidence analysis provides information regarding the certainty of the model and helps identify situations where classification decisions may be ambiguous. Figure 16 presents several examples of confidence distributions generated by the MobileNetV3Large model. For each image, the predicted probabilities associated with the three classes are displayed, allowing a more detailed interpretation of the model's behaviour.

The first example corresponds to a Residential image correctly classified with a confidence of approximately 99.6%. The probability assigned to the remaining classes is nearly zero, indicating that the model is highly confident in its prediction. Similar behaviour can be observed in other residential scenes, where distinctive visual features allow the network to produce highly reliable classifications.

The second example illustrates a City Street image classified correctly with a confidence of approximately 76.1%. Although the prediction remains correct, the model also assigns a non-negligible probability to the Highway category. This behaviour suggests that certain visual characteristics of the scene are shared between the two classes, resulting in a lower level of certainty compared to the previous example.

The analysis of confidence values reveals an important relationship between prediction certainty and classification difficulty. Images containing clear and distinctive scene characteristics generally produce confidence scores above 90%, whereas visually ambiguous scenes often result in lower confidence values. This observation is consistent with the error analysis discussed previously and further supports the conclusion that the primary challenge of the classification task arises from the visual similarity between different road environments.

Overall, confidence analysis provides additional evidence that the proposed model behaves consistently and produces uncertainty estimates that are largely aligned with the complexity of the underlying classification problem.

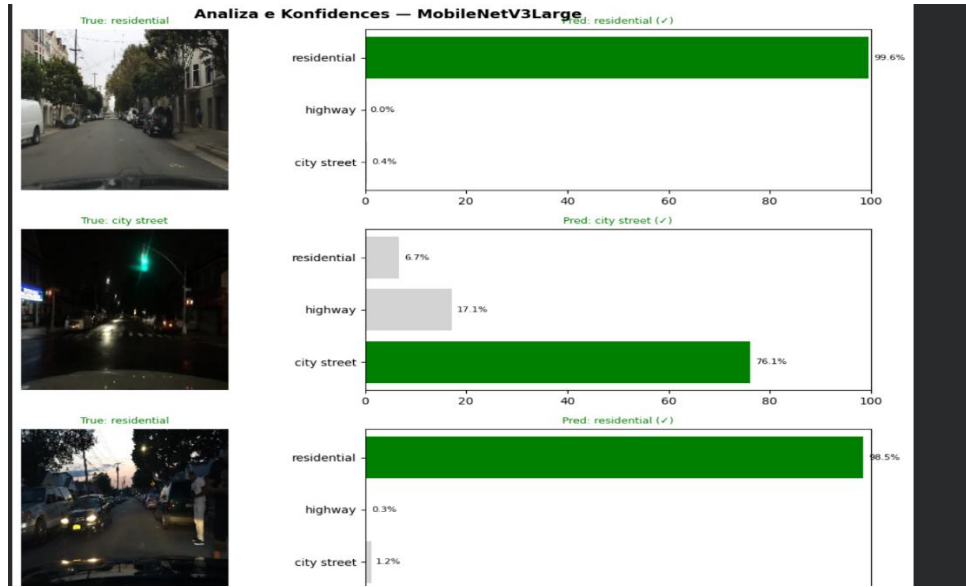


Figure 16: Prediction Confidence Analysis for Correctly Classified Test Images

4.8 Discussion of Results

The experimental results presented throughout this chapter demonstrate the effectiveness of transfer learning for road scene classification tasks. By leveraging pretrained ImageNet representations and adapting them through fine-tuning, the MobileNetV3Large architecture achieved strong performance across all evaluated metrics.

The obtained test accuracy of 75.25% together with a mean AUC value of approximately 0.900 indicates that the model successfully learned meaningful visual representations capable of distinguishing between City Street, Highway and Residential environments. Furthermore, the relatively small gap between training and validation performance suggests that the adopted regularization techniques effectively reduced overfitting and improved generalization capability.

The analysis of confusion matrices, Grad-CAM visualizations and confidence distributions provides additional insight into the behaviour of the model. These analyses reveal that the network focuses on relevant road-related structures when generating predictions and that most classification errors arise from genuine visual similarities between scene categories rather than arbitrary decision-making processes.

Among the three classes, Highway achieved the highest classification performance, indicating that highway environments possess distinctive visual characteristics that are easier to recognize. In contrast, City Street proved to be the most challenging category due to its greater visual diversity and overlap with the other scene types. Nevertheless, the model

maintained satisfactory performance across all categories and produced balanced precision, recall and F1-score values.

Although the obtained results are encouraging, several opportunities for future improvement remain. Additional training data, more advanced data augmentation techniques, semantic segmentation information and larger transformer-based architectures could potentially further improve classification accuracy. Nevertheless, the results obtained in this project demonstrate that modern convolutional neural networks combined with transfer learning constitute an effective solution for automated road scene understanding and classification tasks.

4.9 Comparative Analysis of Deep Learning Architectures

4.9.1 MobileNetV3Large with Transfer Learning

In order to better understand the impact of Transfer Learning, an additional experiment was performed using the MobileNetV3Large architecture without pretrained ImageNet weights. In this case, the model was initialized with random parameters and trained only on the selected dataset. The objective was to compare the performance of a model trained from scratch with the performance obtained using Transfer Learning.

The obtained results were significantly lower than those achieved with the pretrained version. The model reached a test accuracy of only 33.33%, which is equivalent to random performance for a balanced three-class classification problem. The classification report showed that the network failed to learn meaningful visual features and classified almost every image as the *highway* class.

The confusion matrix presented in Figure confirms this behavior. All *city street* and *residential* images were incorrectly classified as *highway*, while only the *highway* class achieved correct predictions. This indicates that the network converged to a trivial solution instead of learning discriminative representations for each scene category.

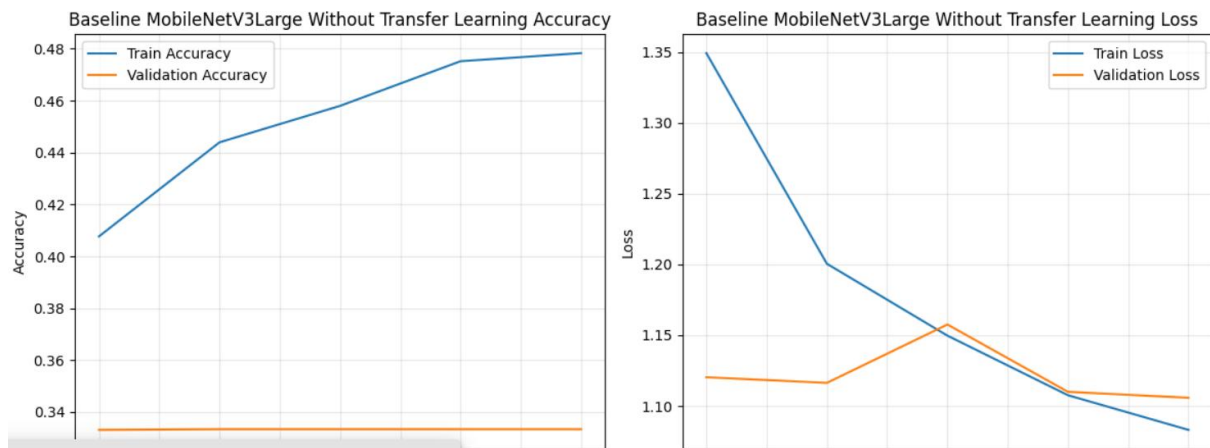


Figure 17: ROC Curves for MobileNetV3Large Without Transfer Learning

The ROC curves shown in Figure further demonstrate the poor classification capability of the model. The obtained AUC values remained close to random classification, especially for the *residential* class. Compared to the Transfer Learning approach, the discriminative power of the network was considerably weaker.

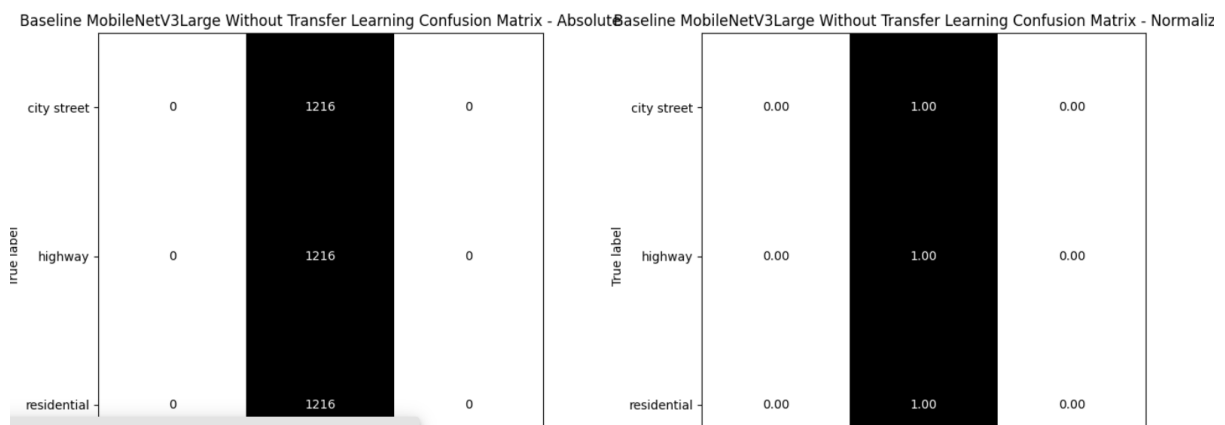


Figure 18: Confusion Matrix for MobileNetV3Large Without Transfer Learning

These results clearly highlight the importance of Transfer Learning in image classification tasks. Pretrained ImageNet weights provide useful low-level and high-level visual features that allow the network to converge faster and achieve significantly better performance. Without Transfer Learning, the model was unable to learn robust representations from the available training data, resulting in poor classification accuracy and unstable predictions.

Model	Accuracy	mAUC
MobileNetV3Large (Without Transfer Learning)	33.33%	0.55

MobileNetV3Large (With Transfer Learning)	75.25%	0.90
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4.9.2 EfficientNetV2

To further evaluate the effectiveness of transfer learning for road scene classification, the EfficientNetV2S architecture was also trained and tested on the same dataset. EfficientNetV2S is a modern convolutional neural network designed to achieve high classification performance while maintaining computational efficiency. Similar to MobileNetV3Large, pretrained ImageNet weights were used as the starting point of the training process. Figure presents the training and validation accuracy and loss curves obtained during training. Both accuracy curves increased steadily throughout the epochs, while the corresponding loss values decreased continuously. The validation accuracy remained consistently higher than the training accuracy, indicating stable learning behavior and good generalization capabilities.

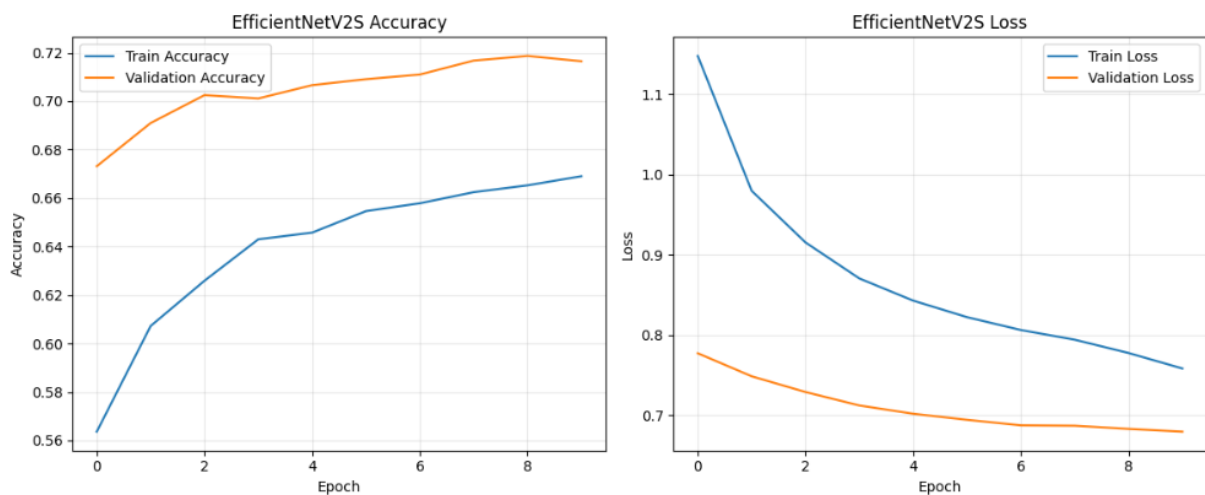


Figure 19: Training and validation accuracy and loss curves of EfficientNetV2S.

The final model achieved a test accuracy of 72.15% with a test loss of 0.6578. The highest class accuracy was obtained for the highway category (79.61%), followed by residential (75.66%) and city street (61.18%). These results demonstrate that the model was able to successfully distinguish between different road environments, although city street scenes remained the most challenging category. The confusion matrices shown provide a more detailed view of the classification performance. The model correctly classified 968 highway images and 920 residential images, while city street images produced the highest number of

misclassifications. Most errors occurred between city street and residential scenes due to similarities in road structure, buildings, and surrounding urban elements.

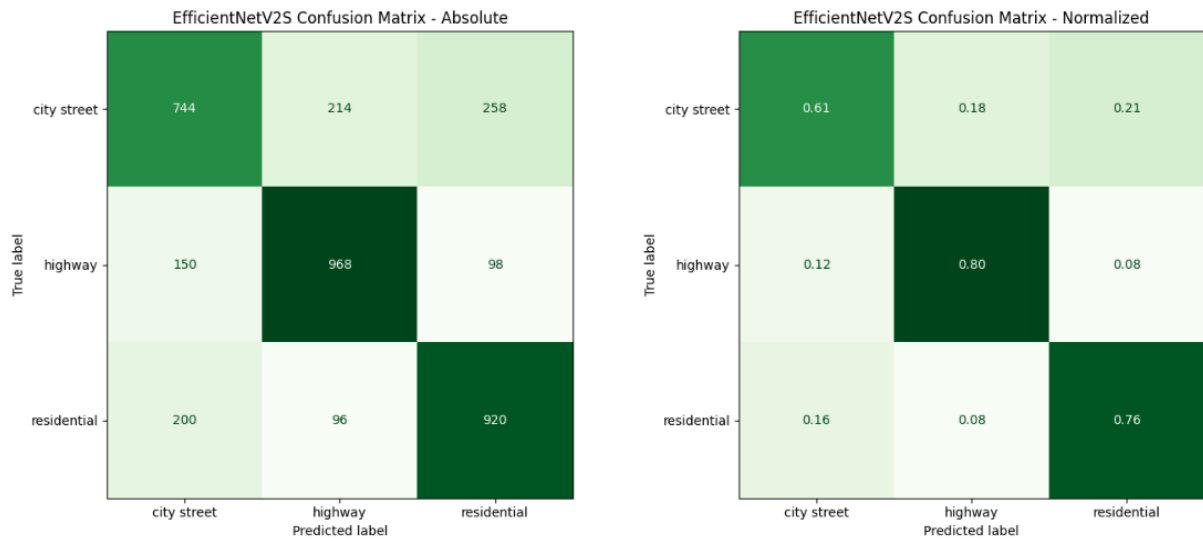


Figure 20: Confusion Matrix obtained by EfficientNetV2S.

Figure above illustrates the ROC curves generated for the three scene categories. The obtained AUC values were 0.829 for city street, 0.917 for highway, and 0.893 for residential, resulting in an overall mean AUC of approximately 0.880. These values confirm that EfficientNetV2S achieved strong discriminative performance across all classes.

Overall, EfficientNetV2S produced competitive classification results and demonstrated strong generalization capabilities. Although its performance was slightly lower than the best MobileNetV3Large configuration, the model achieved robust classification accuracy and reliable class separation, making it a suitable architecture for road scene recognition tasks.

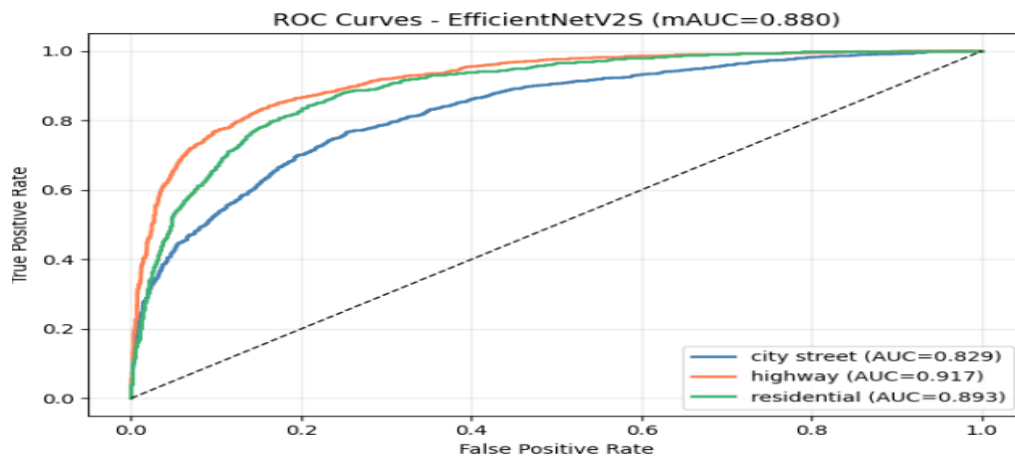


Figure 21: ROC curves of EfficientNetV2S

4.9.2 ConvNeXtSmall

ConvNeXtSmall was evaluated as an additional deep learning architecture in order to compare its performance with MobileNetV3Large and EfficientNetV2S. The model demonstrated a strong classification capability across all three scene categories and produced competitive results on the test dataset.

The confusion matrix analysis shows that the Highway class achieved the highest classification accuracy, with approximately 81% of highway images correctly recognized. The City Street category reached 70% accuracy, while the Residential category achieved 69%. Most of the classification errors occurred between Residential and City Street scenes, which is expected because both categories often contain similar visual elements such as buildings, parked vehicles, intersections, and urban road structures. Despite these challenges, the model maintained a balanced performance across all classes without showing a strong bias toward a single category.



Figure 22: Confusion Matrix for EfficientNetV2S

A closer examination of the normalized confusion matrix indicates that ConvNeXtSmall was able to distinguish highway environments particularly well, with only a small percentage of highway images being confused with the other classes. The Residential class presented the greatest difficulty, mainly due to visual similarities with urban streets. Nevertheless, the model was still able to correctly classify the majority of residential scenes.

The ROC curve analysis further confirms the effectiveness of the model. ConvNeXtSmall achieved a mean Area Under the Curve (mAUC) of 0.894, indicating a high overall discriminative capability. The highest AUC value was obtained for the Highway class (0.924), followed by Residential (0.908) and City Street (0.849). All ROC curves remain

significantly above the random classification baseline, demonstrating that the model learned meaningful visual representations of the different road environments.

When compared with the other architectures evaluated in this project, ConvNeXtSmall achieved a performance level very close to MobileNetV3Large and slightly higher than EfficientNetV2S in terms of mean AUC. These results suggest that ConvNeXtSmall is a robust architecture for road scene classification and represents a strong alternative for transfer learning applications in intelligent transportation systems and autonomous driving environments.

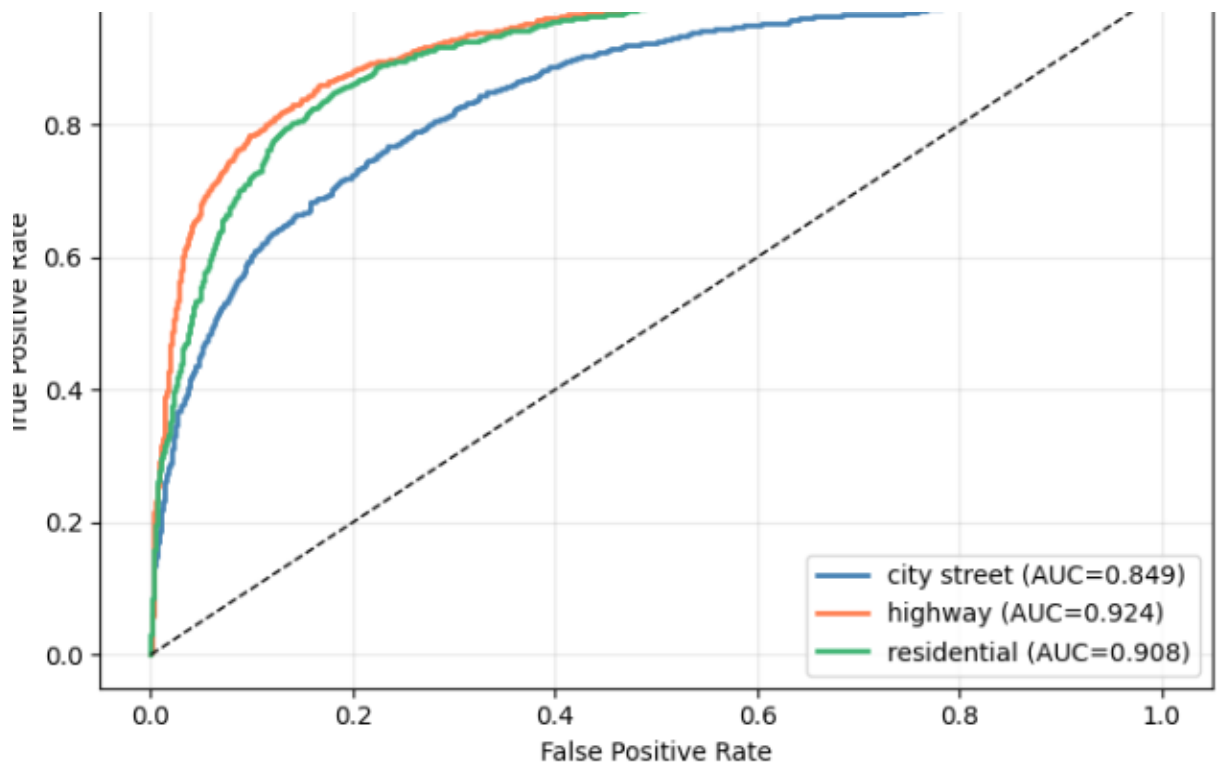


Figure 23: ROC curves obtained by ConvNeXtSmall

4.9.3 Final Architecture Ranking

The final comparison of the evaluated deep learning architectures is presented in this project. This comparison summarizes the performance of all classification approaches implemented during the project and provides a clear overview of the effectiveness of transfer learning for road scene classification. Four different experiments were considered in the final evaluation: MobileNetV3Large with fine-tuning, ConvNeXtSmall, EfficientNetV2S, and a baseline MobileNetV3Large model trained without transfer learning.

The obtained results show a significant difference between the transfer learning approaches and the baseline model. MobileNetV3Large with fine-tuning achieved the highest

overall performance, obtaining a test accuracy of 75.08% and a mean AUC value of 0.900. These results indicate that the model was able to learn robust and discriminative features from the BDD100K dataset and successfully distinguish between the City Street, Highway, and Residential categories. Furthermore, the confusion matrix analysis demonstrated a balanced classification performance across all classes, while the ROC curves confirmed strong discriminative capability.

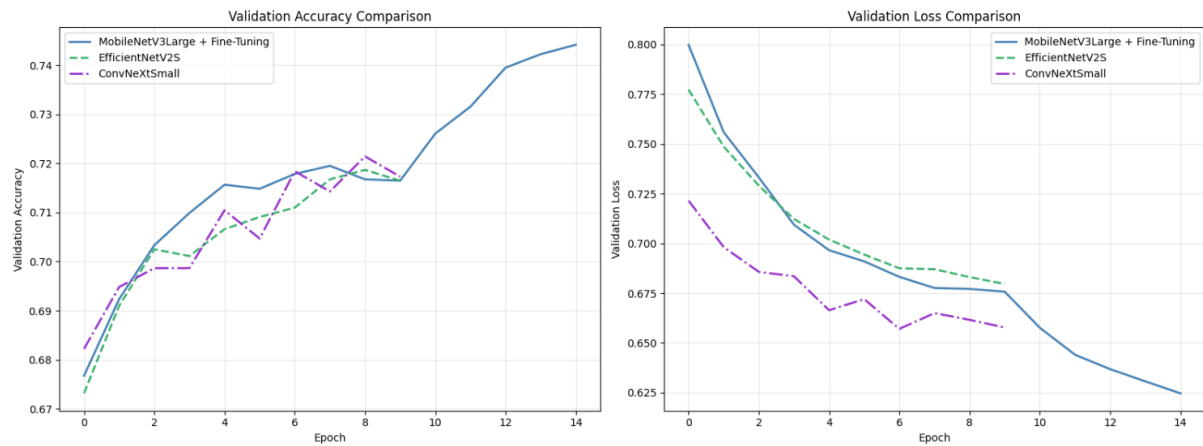


Figure 24: Validation accuracy and validation loss comparison among all models

ConvNeXtSmall achieved the second-best performance with a test accuracy of 73.52% and a mean AUC of 0.894. The difference compared to MobileNetV3Large was relatively small, indicating that ConvNeXtSmall is also a highly effective architecture for this task. The model performed particularly well in the Highway category and demonstrated strong generalization abilities throughout the validation and testing phases. Although its final accuracy was slightly lower than MobileNetV3Large, the results show that ConvNeXtSmall represents a competitive alternative for road scene classification problems.

EfficientNetV2S also produced satisfactory results, reaching a test accuracy of 72.15% and a mean AUC of 0.880. The model achieved stable convergence during training and maintained consistent classification performance across the three scene categories. However, when compared with MobileNetV3Large and ConvNeXtSmall, EfficientNetV2S exhibited slightly lower classification accuracy and a higher degree of confusion between certain scene classes. Despite this, the model still demonstrated the effectiveness of transfer learning and provided strong overall performance.

The baseline experiment without transfer learning obtained substantially lower results, achieving only 33.33% test accuracy and a mean AUC of 0.551. The confusion matrix revealed that the model failed to learn meaningful feature representations and tended to

classify most images into a single category. This experiment clearly demonstrates the limitations of training a deep neural network from scratch on this task and highlights the importance of leveraging pretrained models. The large performance gap between the baseline model and the transfer learning architectures confirms that pretrained feature extraction plays a crucial role in achieving high classification accuracy.

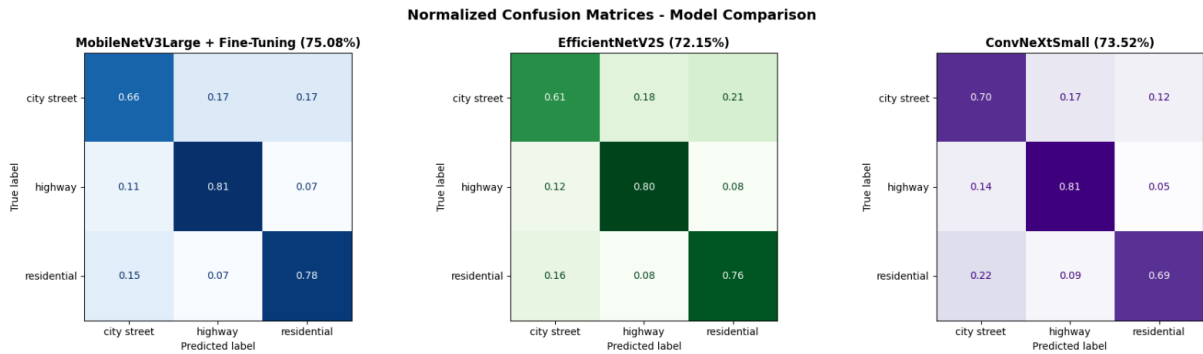


Figure 25: Normalized confusion matrices for all models

From a broader perspective, the comparative analysis shows that transfer learning significantly improves model performance while reducing training complexity and improving generalization. All three transfer learning architectures consistently outperformed the baseline model across every evaluation metric considered in this project. The validation accuracy curves, confusion matrices, ROC curves, and final test results all support the same conclusion.

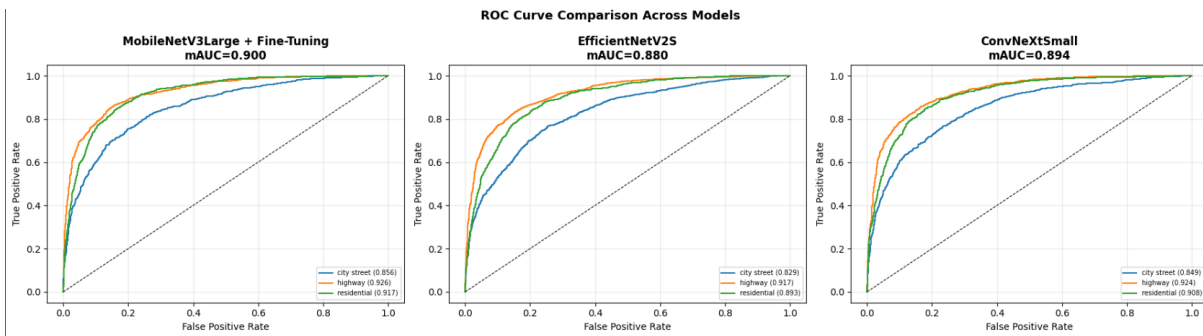


Figure 26: ROC curve for all models

Based on the complete experimental evaluation, MobileNetV3Large was selected as the best-performing architecture for the road scene classification task. It achieved the highest classification accuracy, the highest mean AUC, and the most balanced performance across all categories. The model also demonstrated stable convergence during training and strong generalization on unseen test data. Therefore, MobileNetV3Large can be considered the most suitable architecture among the evaluated approaches for road scene recognition and

intelligent transportation applications. The results obtained throughout this study confirm the effectiveness of transfer learning and demonstrate the potential of modern deep learning architectures for solving real-world computer vision problems.

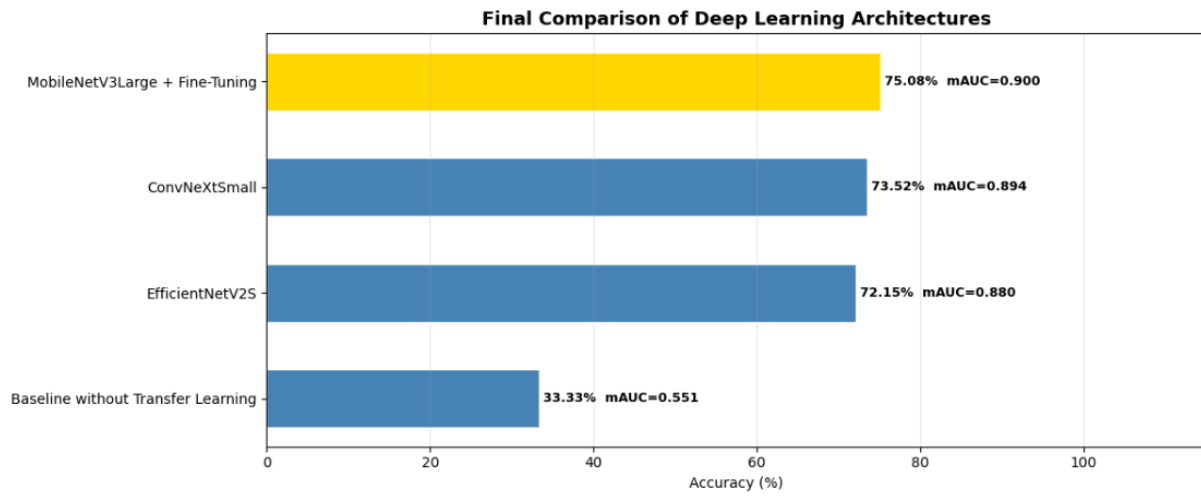


Figure 27: Final Comparison of Deep Learning Architectures

Chapter V: Segmentation

In addition to the road scene classification task, a semantic segmentation approach was implemented in order to identify and extract road regions from aerial images. While classification assigns a single label to an entire image, segmentation performs a pixel-level

analysis, allowing the model to determine which pixels belong to the road class and which belong to the background.

The objective of this part of the project was to evaluate whether a deep learning model could accurately detect road structures from satellite imagery. This task is particularly important in applications such as autonomous navigation, urban planning, traffic monitoring, and geographic information systems, where precise localization of roads is required.

To accomplish this objective, a U-Net architecture was implemented and trained using a road segmentation dataset containing satellite images and their corresponding ground-truth masks. The model learned to predict road regions directly from the input images by comparing its predictions with manually annotated segmentation masks during training.

5.1 Dataset Description

The segmentation dataset consists of satellite images together with binary masks representing road locations. Each image is paired with a corresponding ground-truth mask, where road pixels are represented by white values and background pixels are represented by black values.

The dataset contains a variety of road structures, including:

- Main roads
- Secondary roads
- Urban street networks
- Rural road segments
- Curved and intersecting roads

The diversity of the dataset allows the model to learn different road patterns under varying environmental conditions and geographical layouts.

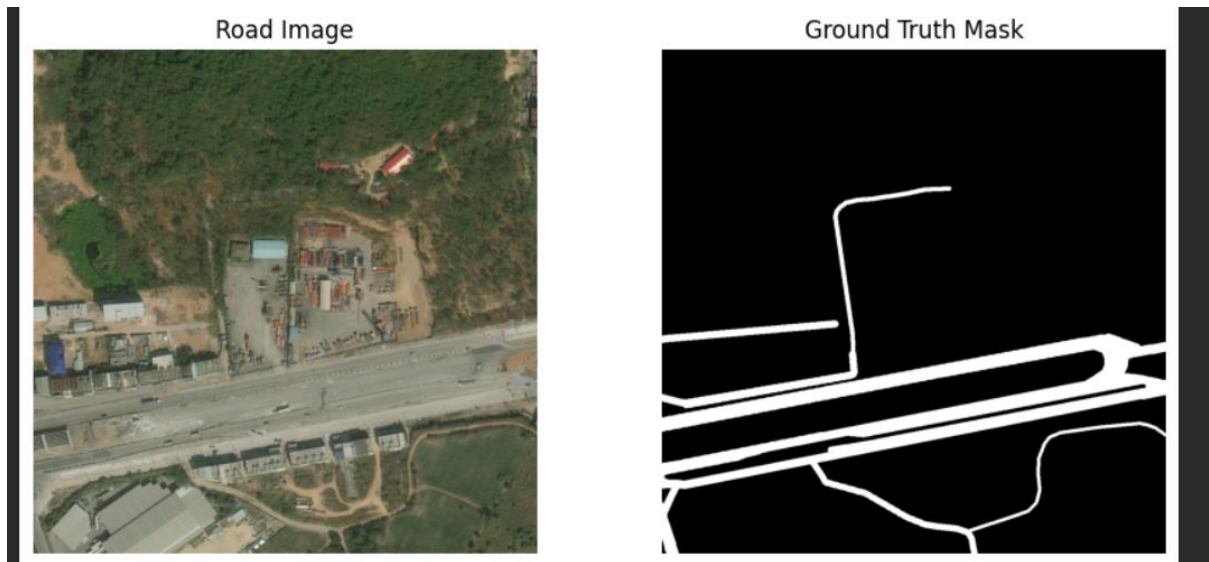


Figure 28: Example of a satellite image and its corresponding ground-truth road mask

5.2 Data Preprocessing

Before training the segmentation model, several preprocessing operations were applied to both the satellite images and their corresponding segmentation masks. These preprocessing steps were essential for ensuring data consistency, improving training efficiency, and enhancing the model's ability to generalize to unseen images.

The first step involved resizing all images and masks to a fixed resolution of 256×256 pixels. Since the original dataset contained images with varying dimensions, resizing was necessary to provide a uniform input size for the neural network. Maintaining a consistent image size simplifies batch processing during training and reduces computational complexity while preserving the most important visual information related to road structures.

After resizing, the RGB satellite images were normalized by scaling pixel values from the original range of 0–255 to the range $[0,1]$. This was achieved by dividing each pixel value by 255. Normalization is a common preprocessing technique in deep learning because it helps stabilize the learning process, accelerates convergence, and prevents large numerical values from negatively affecting gradient updates during optimization.

The segmentation masks also underwent preprocessing before being used for training. Since the objective of the model was to distinguish road pixels from background pixels, the masks were converted into binary images. Pixels representing roads were assigned a value of 1, while all other pixels were assigned a value of 0. This binary representation simplifies the segmentation task and allows the model to focus exclusively on identifying the target class.

Furthermore, binary masks are compatible with segmentation-specific loss functions and evaluation metrics such as Dice Coefficient and Intersection over Union (IoU).

An additional verification step was performed to ensure that each image was correctly paired with its corresponding mask. Accurate alignment between images and masks is crucial in semantic segmentation because any mismatch could negatively affect the learning process and reduce model performance.

To improve model generalization and reduce the risk of overfitting, data augmentation techniques were applied during training. Data augmentation artificially increases the diversity of the training dataset by generating modified versions of existing images while preserving their semantic content. This approach enables the model to learn more robust and invariant feature representations, making it better suited for real-world scenarios where image conditions may vary.

Several augmentation techniques were employed, including horizontal flipping and random brightness adjustments. Horizontal flipping allows the model to observe road structures from different orientations, while brightness variations simulate changes in illumination conditions caused by weather, shadows, or differences in image acquisition. These transformations help the model become less sensitive to visual variations and improve its ability to detect roads under diverse environmental conditions.

The augmentation operations were applied dynamically during training, meaning that different transformed versions of the same image could be presented to the model across multiple epochs. As a result, the effective size of the training dataset increased without requiring additional manually annotated data.

Overall, the preprocessing pipeline played a critical role in preparing the dataset for semantic segmentation. By standardizing image dimensions, normalizing pixel values, converting masks into binary representations, and applying data augmentation techniques, the quality and diversity of the training data were significantly improved, contributing to better segmentation performance.

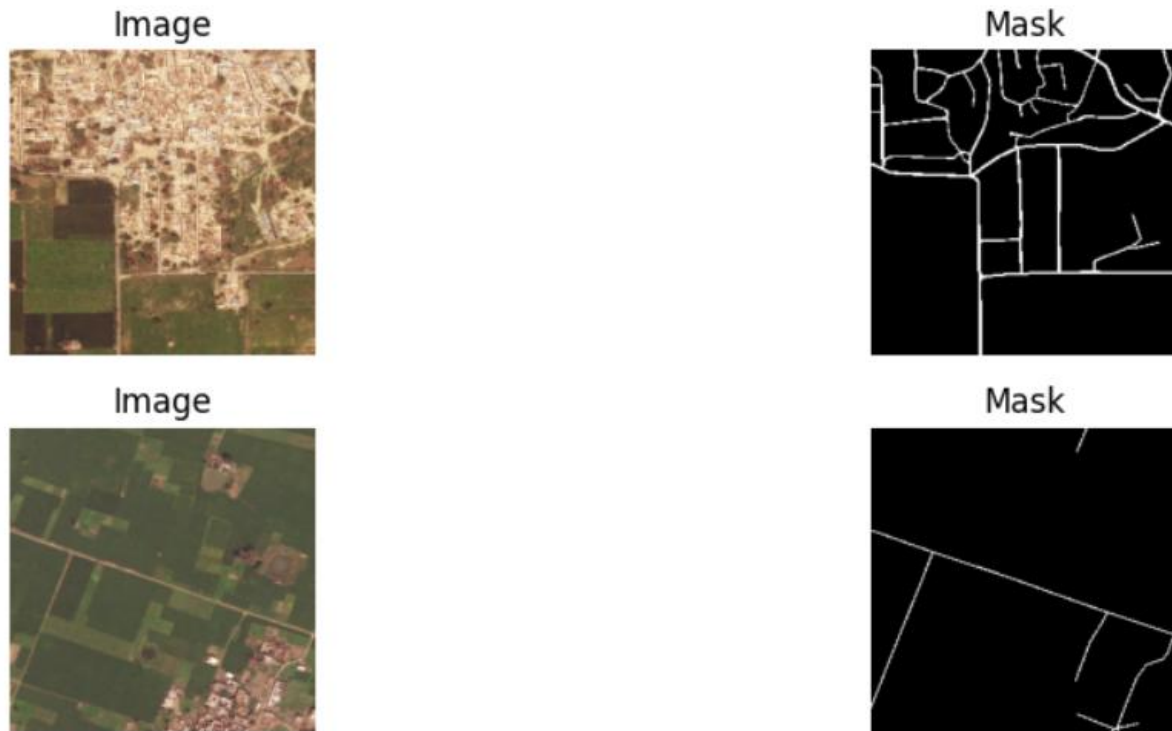


Figure 29: Data Augmentation

5.3 U-Net Architecture

For the segmentation task, the U-Net architecture was selected due to its effectiveness in pixel-level image analysis. U-Net is one of the most widely used architectures for semantic segmentation because it combines accurate localization capabilities with efficient feature extraction. The network follows an encoder-decoder structure. The encoder progressively extracts high-level features from the input image through a sequence of convolutional and pooling layers. During this process, the spatial dimensions are reduced while the number of feature channels increases.

The decoder then reconstructs the segmentation mask by gradually restoring the spatial resolution through upsampling operations. A key characteristic of U-Net is the use of skip connections. These connections transfer feature maps from the encoder directly to the decoder, allowing fine spatial details to be preserved throughout the reconstruction process. This mechanism significantly improves segmentation accuracy, particularly when detecting thin structures such as roads. The final layer produces a probability map indicating the likelihood that each pixel belongs to the road class.

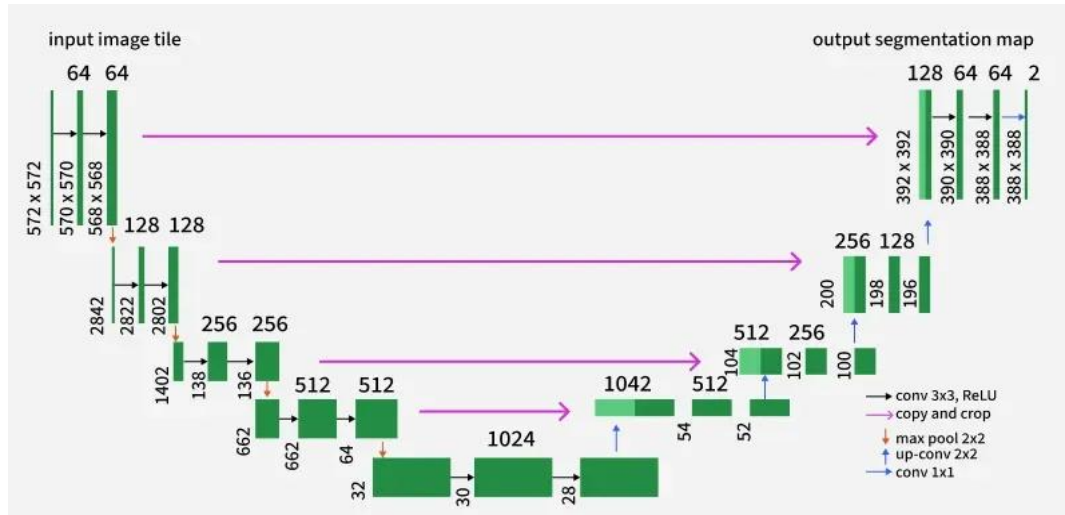


Figure 30: U-Net Architecture

5.4 Training Performance

The model was trained for 10 epochs using the Adam optimizer and a segmentation loss function based on Dice loss. During training, the objective was to maximize the overlap between the predicted road masks and the ground-truth annotations while minimizing segmentation errors.

To evaluate the learning process, three metrics were monitored throughout training:

- Dice Coefficient
- Intersection over Union (IoU)
- Segmentation Loss

These metrics provide complementary information about the model's performance. The Dice coefficient and IoU measure the similarity between predicted masks and ground-truth masks, while the loss function indicates how well the model is optimizing during training.

5.4.1.Dice Coefficient Analysis



Figure 31: Dice Coefficient During Training

Figure illustrates the evolution of the Dice coefficient throughout the training process. At the beginning of training, the Dice score was relatively low because the model had not yet learned meaningful road features from the satellite images. During the first few epochs, a rapid increase can be observed, indicating that the network quickly learned to distinguish road pixels from the background.

As training progressed, the Dice coefficient continued to improve, although at a slower rate. This behavior is expected because the model gradually refines its predictions after learning the most prominent road patterns. Towards the final epochs, the curve begins to stabilize, suggesting that the model is approaching convergence and that additional training would likely result in smaller performance gains. The final Dice coefficient achieved by the model demonstrates a satisfactory level of overlap between the predicted segmentation masks and the ground-truth annotations. This indicates that the network successfully learned the spatial characteristics of road structures present in the dataset.

The figure shows a consistent upward trend, confirming that the segmentation quality improved steadily throughout the training process.

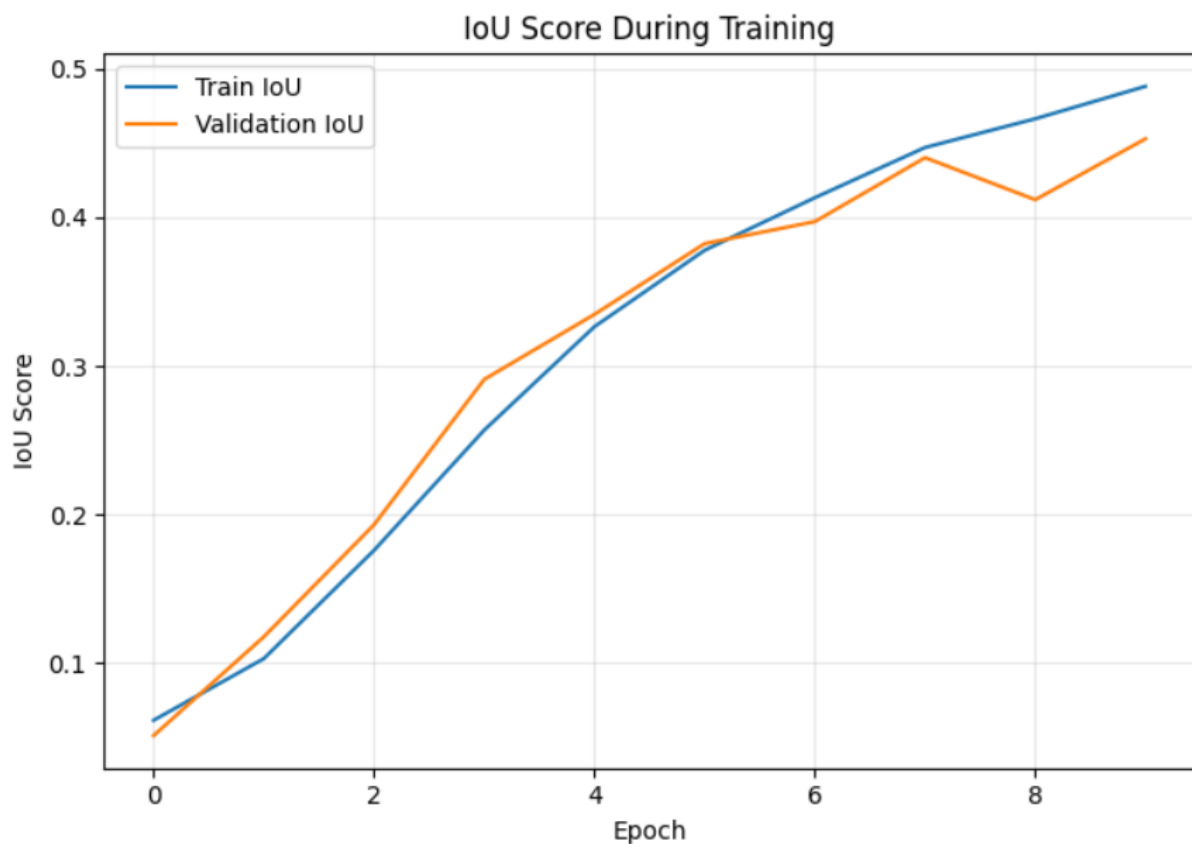


Figure 32:IoU Score During Training

Figure presents the Intersection over Union (IoU) score measured during training. IoU is a stricter evaluation metric than the Dice coefficient because it directly measures the ratio between the overlapping area and the total combined area of the predicted and ground-truth masks.

At the initial epochs, the IoU score was relatively low due to the limited ability of the model to accurately localize road regions. However, as the network learned more representative features, the IoU score increased progressively. The steady upward trend indicates that the predicted masks became increasingly aligned with the actual road annotations.

The improvement observed in the IoU curve confirms that the model was not only identifying roads but also improving the precision of road boundaries. This is particularly important in road segmentation tasks because roads often appear as thin and elongated structures that require accurate localization.

Near the end of training, the IoU curve becomes more stable, suggesting that the model had reached a point where further improvements were incremental rather than substantial.

The figure demonstrates a continuous increase in overlap quality between predicted masks and ground-truth masks, reflecting the model's growing ability to accurately segment road regions.

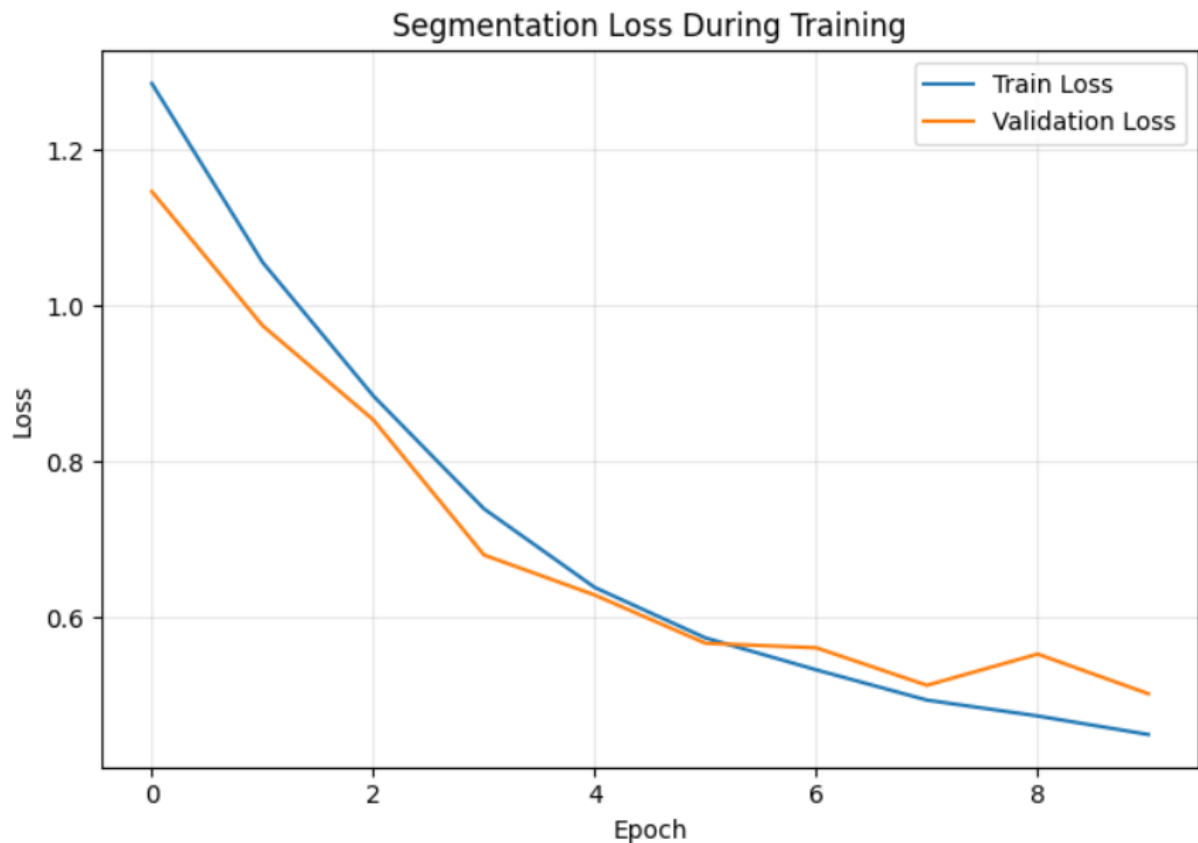


Figure 33: Segmentation Loss During Training

Figure illustrates the behavior of the segmentation loss throughout training. Unlike Dice and IoU, where higher values indicate better performance, lower loss values correspond to improved model predictions.

At the beginning of training, the loss value was relatively high because the model generated inaccurate segmentation masks. During the first epochs, a significant decrease in loss can be observed, indicating that the optimization process was effectively reducing prediction errors.

As training continued, the loss decreased more gradually, reflecting the model's effort to fine-tune its predictions and improve segmentation accuracy. The smooth downward trend

suggests that the learning process remained stable and that no major optimization issues, such as divergence or unstable gradients, occurred during training.

Towards the final epochs, the loss curve begins to flatten, indicating that the model had reached a stable solution and that the optimization process was converging successfully. The decreasing loss curve confirms that the network continuously improved its predictions and successfully learned the relationship between satellite images and their corresponding road masks.

Overall, the three training curves provide strong evidence that the U-Net model learned meaningful road representations from the dataset. The simultaneous increase in Dice coefficient and IoU, combined with the continuous decrease in segmentation loss, demonstrates successful training and effective convergence of the segmentation network.

5.5 Segmentation Results

After completing the training process, the U-Net model was evaluated using a set of previously unseen test images. The purpose of this evaluation was to assess the model's ability to generalize beyond the training data and accurately identify road networks in different geographical environments and image conditions.

The generated segmentation masks demonstrate that the model successfully learned the visual characteristics of roads from satellite imagery. In the majority of test samples, the predicted masks closely matched the corresponding ground-truth annotations, indicating that the network was able to distinguish road surfaces from surrounding elements such as buildings, vegetation, open land, and shadows.

One of the most notable observations is the model's ability to preserve the continuity of road structures. Major roads, intersections, and connected road networks were generally detected with high consistency, allowing the predicted masks to maintain the overall topology of the transportation network. This is particularly important in road extraction tasks, where disconnected predictions can significantly reduce the practical usefulness of the results.

A visual comparison between the original satellite images, the manually annotated masks, and the masks predicted by the model reveals a strong level of agreement. The predicted outputs capture the primary road patterns and successfully follow road trajectories across different regions of the image. In urban areas, where roads often appear close to buildings and other man-made structures, the model was still able to identify the correct road

boundaries with reasonable precision. Similarly, in rural environments, where roads may be narrower and less visually distinct, the network demonstrated the ability to recognize and segment the most prominent road segments.

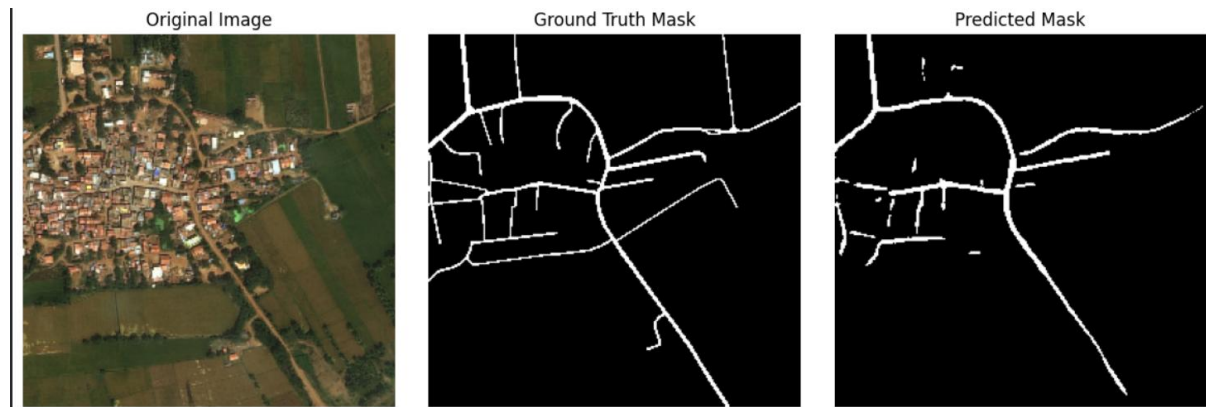


Figure 34: Predicted Mask

Examples of these comparisons are presented through a series of visual results that illustrate the progression from the original satellite image to the corresponding ground-truth annotation and finally to the segmentation mask generated by the model. These examples provide a clear demonstration of how effectively the network learned the road extraction task.

Despite the encouraging performance, several limitations were observed during testing. Thin road segments and very narrow pathways were occasionally only partially detected or completely omitted. This behavior is common in semantic segmentation tasks because small structures occupy relatively few pixels and therefore contribute less information during training. In addition, minor prediction artifacts appeared in isolated regions where image textures resembled road surfaces. Such false-positive detections were generally limited in size and did not significantly affect the overall quality of the segmentation output.

Another challenge involved areas with complex visual patterns, including dense urban zones, shadows cast by buildings, and regions with low image contrast. In these situations, the model sometimes produced slightly irregular boundaries around road edges. Nevertheless, the predicted masks remained sufficiently accurate to preserve the overall road layout and connectivity.

To provide a more intuitive interpretation of the results, an overlay visualization was generated by superimposing the predicted road masks onto the original satellite images. This approach allows the detected road regions to be viewed directly within their geographical

context, making it easier to evaluate the spatial accuracy of the segmentation process. The overlay images clearly highlight the portions of the scene identified as roads and demonstrate how closely the predictions align with the actual road infrastructure visible in the satellite imagery.

The overlay representations also reveal the strengths of the model in capturing long continuous road segments and complex intersections. By combining the original image with the predicted mask, it becomes possible to visually assess both the correctness of the detected roads and the areas where improvements could still be made. These visualizations confirm that the model is capable of extracting meaningful road information from satellite imagery and producing segmentation outputs that are suitable for further analysis and mapping applications.



Figure 35: Road Segmentation Overlay Visualization

Overall, the qualitative evaluation indicates that the proposed U-Net architecture achieved reliable road segmentation performance. The generated masks accurately represent the majority of road structures present in the test images, demonstrating the effectiveness of deep learning techniques for automated road extraction from aerial and satellite data.

5.6 Quantitative Evaluation

The final performance obtained on the test dataset is summarized below:

- Accuracy: 97.11%
- Dice Coefficient: 0.6154
- IoU Score: 0.4487

- Loss: 0.4959

The high accuracy value indicates that the majority of pixels were classified correctly. However, in segmentation problems, Dice coefficient and IoU are generally considered more informative metrics because they directly measure the overlap between predicted masks and ground-truth annotations. This is particularly important in road extraction tasks, where road pixels typically represent only a small portion of the entire image, making accuracy alone insufficient for evaluating model performance. The obtained Dice score of 0.6154 and IoU score of 0.4487 demonstrate that the model successfully learned the road extraction task and was able to generate meaningful segmentation masks on previously unseen images. These values indicate a reasonable level of agreement between the predicted masks and the ground-truth annotations, showing that the model was capable of identifying the main road structures while maintaining continuity across large segments.

The loss value of 0.4959 further confirms that the network converged during training and achieved stable performance on the test dataset. Although some road segments were occasionally missed and a small number of false-positive predictions were observed, the overall segmentation quality remained satisfactory across different image conditions and road layouts.

Overall, the results confirm that the proposed U-Net architecture provides an effective solution for road segmentation using satellite imagery. The combination of high pixel-wise accuracy, moderate Dice and IoU scores, and visually consistent segmentation outputs demonstrates the capability of the model to extract road networks from complex satellite scenes and highlights its potential for practical remote sensing applications.